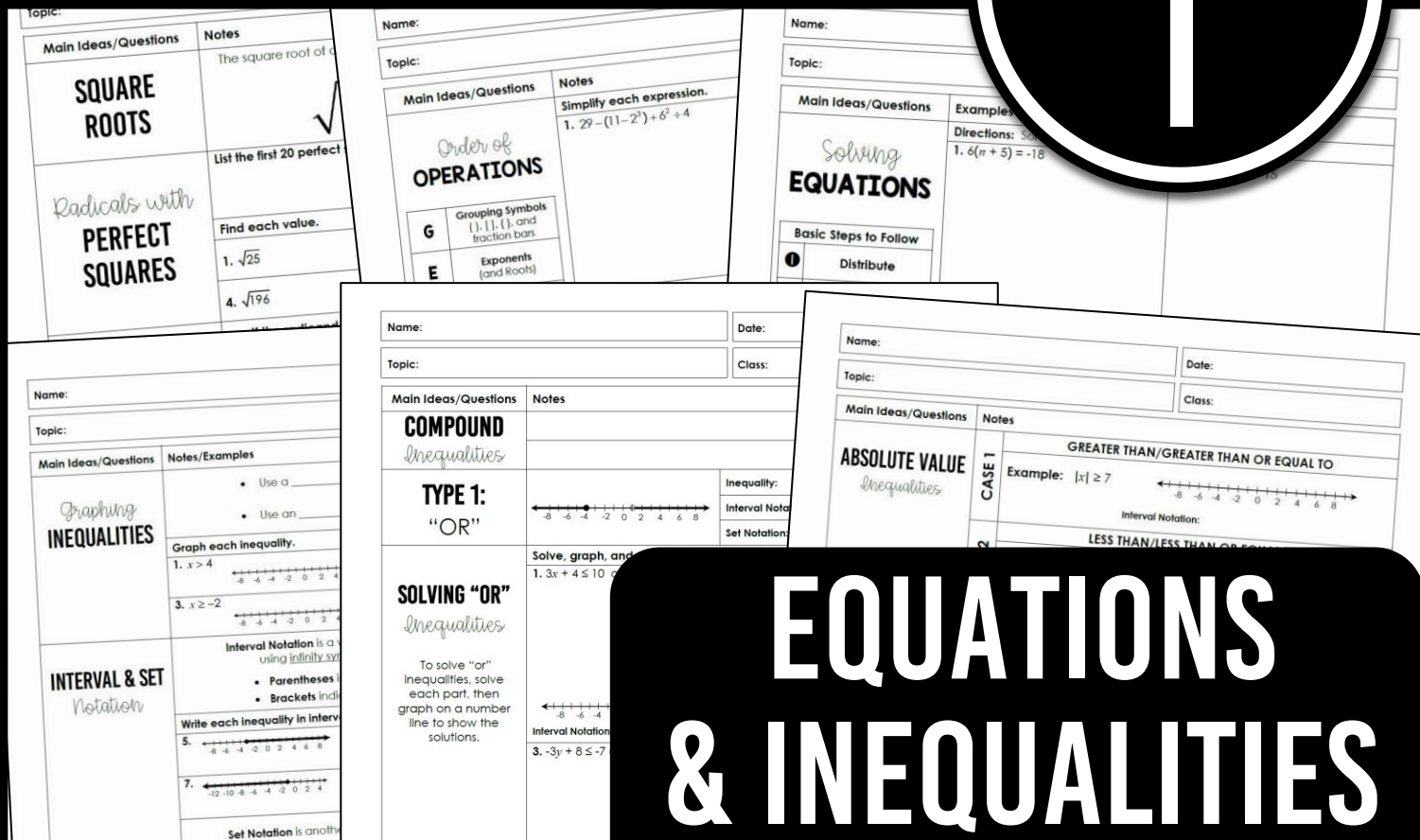


ALGEBRA 2

Unit 1



EQUATIONS & INEQUALITIES

NOTES • HOMEWORK • QUIZZES • TEST

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Unit 1 - Equations & Inequalities: Sample Unit Outline

	TOPIC	HOMEWORK
DAY 1	Simplifying Radicals & Classifying Numbers (Real Number System)	HW #1
DAY 2	Expressions & Operations: Order of Operations, Evaluating Expressions, Simplifying Expressions	HW #2
DAY 3	Solving Equations Review: Multi-Step Equations, Proportions, Literal Equations	HW #3
DAY 4	Multi-Step Equation Word Problems	↓
DAY 5	Absolute Value Equations	HW #4
DAY 6	Quiz 1-1	None
DAY 7	Graphing Inequalities, Interval & Set Notation, Solving Multi-Step Inequalities	HW #5
DAY 8	Compound Inequalities	HW #6
DAY 9	Special Cases of Compound Inequalities	↓
DAY 10	Absolute Value Inequalities	HW #7
DAY 11	Quiz 1-2	None
DAY 12	Unit 1 Review	Study for Test
DAY 13	UNIT 1 TEST	None

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes	
SQUARE ROOTS	The square root of a number, a , can be written as a radical expression :	
	<i>Radicals with</i> PERFECT SQUARES	
SIMPLIFYING <i>Radicals</i>	List the first 20 perfect squares:	
	Find each value.	
	1. $\sqrt{25}$	2. $\sqrt{256}$
	3. $-\sqrt{121}$	
	4. $\sqrt{196}$	5. $-\sqrt{324}$
	6. $\sqrt{\frac{9}{100}}$	
	If the radicand is NOT a perfect square, we can simplify the radical.	
	①	Find the largest perfect square that the radicand is divisible by. Break down the radical using this number.
	②	Take the square root of the perfect square. Take it out of the radical.
	③	Leave the "leftover" under the radical symbol.
	Simplify each expression	
	7. $\sqrt{44}$	8. $\sqrt{288}$
	9. $\sqrt{147}$	10. $-\sqrt{80}$
	11. $\sqrt{600}$	12. $\sqrt{162}$
	13. $-\sqrt{75}$	14. $\sqrt{448}$

REAL NUMBERS ()	The Real Numbers consist of the following sets of numbers: • Irrational Numbers: Numbers in which the decimal form never terminates and does not repeat. Common examples are π and non-perfect square roots. • Rational Numbers: Numbers in which the decimal form either terminates or repeats. Rational numbers can always be written as a fraction (a/b) where a and b are integers. Subsets of Rational Numbers: ○ Integers: $\{..., -3, -2, -1, 0, 1, 2, 3, ...\}$ ○ Whole Numbers: $\{0, 1, 2, 3, 4, ...\}$ ○ Natural Numbers: $\{1, 2, 3, 4, 5, ...\}$		
ORGANIZING <i>the Real Numbers</i>	Set	Letter	Examples
	Irrational Numbers		
	Rational Numbers		
	Integers		
	Whole Numbers		
	Natural Numbers		
CLASSIFYING <i>Numbers</i>	Name all sets to which each value belongs.		
	15. -10	16. $\sqrt{30}$	17. $1\frac{7}{9}$
	18. $\frac{24}{8}$	19. 2.48	20. $-\sqrt{81}$
	21. $\pi - \pi$	22. $\sqrt{\frac{16}{25}}$	23. $0.\overline{15}$
	24. 109	25. $-\frac{8}{3}$	26. $\frac{\sqrt{7}}{\sqrt{7}}$

Name: _____

Unit 1: Equations & Inequalities

Date: _____ Per: _____

Homework 1: Simplifying Radicals & Classifying Numbers

Simplify each expression.

1. $\sqrt{64}$

2. $-\sqrt{225}$

3. $\sqrt{400}$

4. $\sqrt{27}$

5. $\sqrt{363}$

6. $-\sqrt{216}$

7. $\sqrt{125}$

8. $-\sqrt{32}$

9. $\sqrt{245}$

10. $\sqrt{96}$

11. $\sqrt{392}$

12. $-\sqrt{192}$

Name ALL SETS to which each number belongs.

13. $-\frac{5}{3}$

14. $\sqrt{49}$

15. $0.\bar{6}$

16. π

17. $-\frac{36}{4}$

18. $\sqrt{90}$

19. 1.125

20. $-\sqrt{169}$

21. $\frac{4}{11}$

22. Place the LETTER of each value on the diagram in the smallest subset that contains the value.

A. $-\sqrt{196}$

B. $\frac{\pi}{\pi}$

C. $2\frac{6}{7}$

D. $\sqrt{17}$

E. $1.\bar{83}$

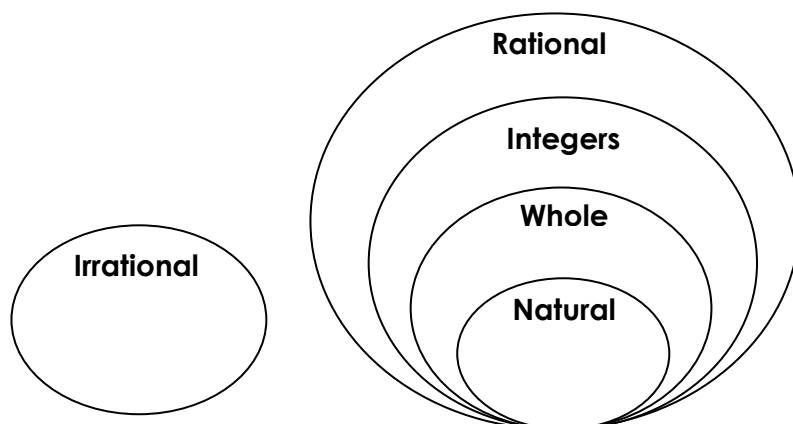
F. -0.75

G. $\sqrt{2} - \sqrt{2}$

H. $\sqrt{\frac{1}{4}}$

I. $\frac{35}{7}$

J. $\frac{1}{13}$



Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes									
Order of OPERATIONS <table><tr><td>G</td><td>Grouping Symbols (), [], { }, and fraction bars</td></tr><tr><td>E</td><td>Exponents (and Roots)</td></tr><tr><td>MD</td><td>Multiplication & Division (left → right)</td></tr><tr><td>AS</td><td>Addition & Subtraction (left → right)</td></tr></table>	G	Grouping Symbols (), [], { }, and fraction bars	E	Exponents (and Roots)	MD	Multiplication & Division (left → right)	AS	Addition & Subtraction (left → right)	Simplify each expression.	
	G	Grouping Symbols (), [], { }, and fraction bars								
	E	Exponents (and Roots)								
	MD	Multiplication & Division (left → right)								
	AS	Addition & Subtraction (left → right)								
1. $29 - (11 - 2^3) + 6^2 \div 4$		2. $4^3 - -8 \cdot 6 \div 16$								
3. $\frac{7 - (10 - 12 + 5^2)}{9 - 2^2}$		4. $\frac{(2 - 3^2)^2 + \sqrt{81}}{10 - 6 \div 3}$								
5. $\frac{[15 - (-6 + 2 + \sqrt{9})]^2}{3 - 3^3}$		6. $\frac{(9 - 15)^2 \cdot (7 - 2^2)}{\sqrt{49} + 5} - -5 $								
EVALUATING Expressions	Evaluate each expression given the variable replacements.									
	7. $\frac{a \cdot b^2}{c + 12}$ (if $a = 3$, $b = -4$, and $c = 8$)	8. $ 10m - 8n^2 $ (if $m = 2$ and $n = -3$)								

	9. $-x^2 + 7(y + 2) - 1$ (if $x = 5$ and $y = -9$)	10. $3r - 4s$ (if $r = 5$ and $s = \frac{2}{3}$)
	11. $ x^2 - y^2 - x \cdot y$ (if $x = -6$ and $y = 8$)	12. $\frac{2}{5}(a + 2b)^2$ (if $a = 3$ and $b = -4$)
	13. $\frac{w^3 - 4v}{\sqrt{wv}}$ (if $w = 2$ and $v = 18$)	14. $2m^2 + 9m$ (if $m = \frac{1}{4}$)
SIMPLIFYING Expressions	Simplify each expression by distributing and/or combining like terms.	
	15. $16x - 11 - 7x + 10$	16. $13 + 2(3n - 7) + 19n$
	17. $6(c - 2) - (8c + 1)$	18. $-15y - 7z + y + 9z$
	19. $3(3a - 5b) - (a - 7b)$	20. $-8(2x + y) - 3(-5x - 2y)$

Name: _____

Unit 1: Equations & Inequalities

Date: _____ Per: _____

Homework 2: Expressions & Operations

**** This is a 2-page document! ******Simplify each expression.**

1. $25 - (\sqrt{16} - 1) \cdot (3 - 9)^2$

2. $|5 - 2(9)| + 9^2 \div 3$

3. $\frac{2^5 - 4 \cdot 3^3}{7 + (1 - \sqrt{100})}$

4. $\frac{(\sqrt{225} - 11) \cdot 12}{-12 - (8 - 6^2)} - |-3|$

5. $\frac{3}{2}[(58 - 10^2) \div (\sqrt{16} + 3)]$

6. $\frac{56 \div (7 - 9)^3 - 25}{24 - 5 \cdot 4}$

Evaluate each expression given the replacement values.

7. $a^2 + 2(b - 6) - 17$ (if $a = -7$ and $b = 2$)

8. $\frac{10xy}{8x - 2y + 10}$ (if $x = 4$ and $y = -7$)

9. $\frac{3}{4} m - 2n^2$ (if $m = -4$ and $n = 6$)	10. $\frac{(4c + 9d)^2}{c^2 - 3d^2}$ (if $c = 3$ and $d = -2$)
11. $-w^2 - 11uv + 2v^2$ (if $u = -1$, $v = 8$, and $w = -3$)	12. $\frac{2}{5}(x - 4y) + \frac{1}{2}(2x + y)$ (if $x = 1$ and $y = -6$)
13. $\frac{a^3 - \sqrt{2ab}}{(a - b)^2}$ (if $a = 4$ and $b = 8$)	14. $p^2 + 2pq - 3q^2$ (if $p = \frac{1}{4}$ and $q = \frac{2}{3}$)
Simplify each expression below by distributing and/or combining like terms.	
15. $9x - 2 + 3x - 5$	16. $-2(6m - 7) + 3(m - 13)$
17. $17 - (2w + 3) + 7(10 - w)$	18. $19x + 2y - 10y - 7x$
19. $3(-2a + 8b) - (9b - a)$	20. $7x(5 - y) - 3(9x + 2) - 4xy$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Examples	
Solving EQUATIONS Basic Steps to Follow 1 Distribute 2 Simplify Each Side by combining like terms. 3 Move Variables to one side. 4 Solve using inverse operations.	Directions: Solve each equation below.	
	1. $6(n + 5) = -18$	2. $\frac{3(5 - 3x)}{-4} = -15$
	3. $2(2 - 7y) - 5 = -29$	4. $-16 = -\frac{15}{16}\left(\frac{4}{5}q + \frac{32}{3}\right)$
	5. $10a - 7 = 21 - 6a$	6. $\frac{5}{6}w + 21 = -\frac{1}{3}(2w - 9)$
	7. $-5(8p + 6) = 5 - 5p$	8. $16 - (7m + 3) = 8(1 - m)$

	9. $\frac{10x+17-(x-9)}{2} = 3x+1$	10. $\frac{9}{8}\left(4-\frac{1}{2}a\right) = 3\left(\frac{1}{4}a+\frac{5}{2}\right)$
SPECIAL <i>Cases</i>	11. $1+3c = 5(1-c) + 8c$	12. $40-8x = -8(x-5)$
PROPORTIONS	13. $\frac{5}{2} = \frac{40}{x+4}$	14. $\frac{10}{5x-9} = \frac{8}{2x-8}$
LITERAL <i>Equations</i>	Directions: Solve each equation or formula for the specified variable.	
	15. $C = 2\pi r$ (for π)	16. $A = \frac{1}{2}ap$ (for p)
	17. $\frac{a+b}{c} = 5$ (for b)	18. $3x-3y = 24$ (for x)
	19. $7w = \frac{1}{2}(w+v)$ (for v)	20. $S = (n-2) \cdot 180$ (for n)

USING EQUATIONS *to solve* WORD PROBLEMS

Directions: Set up and solve an equation for each problem below.

- | | |
|--|---|
| <p>1. The product of 7 and a number, increased by 3, is equal to twice the number, subtracted from 39. Find the number.</p> | <p>2. Five less than twice a number is equal to one half the difference of three times the number and 13. Find the number.</p> |
| <p>3. One number is 9 more than twice another number. If the sum of the numbers is 129, find both numbers.</p> | <p>4. Kara's golf score was 1 less than three fourths of Greg's golf score. If the sum of their scores was 153, find Kara's golf score.</p> |
| <p>5. The length of a rectangle is one more than four times its width. If the perimeter of the rectangle is 62 meters, find the dimensions of the rectangle.</p> | <p>6. The length of one base of a trapezoid is 19 less than five times the length of the other base. If the trapezoid has a height of 18 feet and an area of 477 ft^2, find the length of the longer base.</p> |

7. The measure of one angle is six more than half the measure of another angle. If the angles are complementary, find both measures.	8. Kelly, Laura, and Tia sold boxes of cookies for a fundraiser. Kelly sold 15 boxes less than Laura. Tia sold twice as many boxes as Kelly. If they sold 83 boxes of cookies in total, how many boxes did each girl sell?
9. The measure of the vertex angle in an isosceles triangle is six less than four times the measure of a base angle. Find the measure of the vertex angle.	10. The sum of four consecutive integers is 222. Find the integers.
11. The sum of three consecutive odd integers is 93. Find the integers.	12. Find three consecutive even integers such that the sum of the smallest integer and twice the median integer is 20 more than the largest integer.

Name: _____

Unit 1: Equations & Inequalities

Date: _____ Per: _____

Homework 3: Solving Equations

**** This is a 2-page document! ****

Solve each equation.

1. $-5(3x - 4) + 4 = 114$

2. $-3 = \frac{9(5 - 2k)}{3}$

3. $\frac{15}{8}m - 11 = \frac{7}{4}m + 1$

4. $-8\left(3x - \frac{1}{4}\right) = 2(9 - 7x)$

5. $\frac{4}{9}\left(\frac{3}{8}x - \frac{27}{16}\right) = \frac{13}{4}$

6. $10 - (8p + 3) = 9(2p - 5)$

7. $3(n + 11) - 5n = -2(n - 12) + 9$

8. $-9 - (18v - 6) = 2(7v - 16) - 3v$

9. $-8 - 3(w + 13) = 4(w + 11) - 7w$	10. $\frac{3a - 1}{8} = \frac{a + 9}{4}$
11. $4c + 9 = \frac{8c + 1 - (2c - 11)}{3}$	12. $\frac{5x + 6}{3x - 9} = \frac{7}{3}$
Write and solve an equation to solve the problem.	
13. The measure of one angle is 8 degrees more than one-third the measure of another other angle. If the angles are supplementary, find both angle measures.	14. Find three consecutive odd integers such that the sum of the smallest number and four times the largest number is fifty-one more than twice the median number.
Solve each equation for the indicated variable.	
15. $A = \frac{1}{2}\pi r^2$ (for π)	16. $\frac{x - y}{2z} = 3$ (for x)
17. $4m - n = 1$ (for n)	18. $9a = \frac{12a + 3b}{4}$ (for b)

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes	
WARM-UP	1. If $ x = 7$, what are the possible values of x ?	
	2. If $ x = -4$, what are the possible values of x ?	
<i>Steps to solve</i> ABSOLUTE VALUE EQUATIONS	❶	ISOLATE the absolute value expression.
	❷	CREATE TWO CASES. Set the "inside" equal to both the positive and negative value of the number on the opposite side of the equal sign.
	❸	SOLVE both equations.
	❹	CHECK for extraneous solutions.
What are Extraneous Solutions?		
EXAMPLES	Solve each equation. Be sure to check for extraneous solutions.	
	3. $ x - 9 = 5$	4. $3 = 2k + 7 $
	5. $ y - 1 + 17 = 23$	6. $3 m + 8 = 39$
	7. $16 r - 5 = -32$	8. $\frac{ y - 10 }{-2} = -6$

9. $-5 = |8 - p| - 11$

10. $\frac{|3a + 12|}{9} = 0$

11. $9 - 4|c + 10| = -15$

12. $-16 = -12\left|\frac{k}{5}\right| + 8$

13. $|3x - 9| = x + 1$

14. $2|h + 8| = 14h$

15. $|8 - 2n| - 5n = n + 2$

16. $-4w - 2 = |6w + 20|$

Name: _____

Unit 1: Equations & Inequalities

Date: _____ Per: _____

Homework 4: Absolute Value Equations

**** This is a 2-page document! ****

Solve each equation. Be sure to check for extraneous solutions.

1. $|x - 7| = 16$

2. $|-8k| = 24$

3. $2 = |8p + 10|$

4. $\left|\frac{y + 9}{4}\right| = 5$

5. $-9|c + 8| = -99$

6. $|2s| + 19 = 13$

7. $|2 + 7v| + 11 = 27$

8. $-6|8x - 4| = -72$

9. $53 = 5|p - 4| + 8$

10. $8 - 12|k + 5| = -4$

11. $-51 = 3 - |6 - 12r|$

12. $\frac{|6 - 2m|}{-4} - 7 = -19$

13. $|4x - 1| = 2x + 13$

14. $5n - 3 = |n + 9|$

15. $z - 10 = |7z| - 16$

16. $\frac{|12 - x|}{2} = 4x$

Name: _____

Algebra 2

Date: _____ Per: _____

Unit 1: Equations & Inequalities

Quiz 1-1: Expressions & Equations

Simplify each radical expression.

1. $\sqrt{54}$

2. $-\sqrt{176}$

1. _____

2. _____

3. $\sqrt{300}$

4. $\sqrt{252}$

3. _____

4. _____

Name all sets to which each value belongs. (Use R, I, Q, Z, W, N)

5. $\sqrt{32}$

6. $-\frac{60}{12}$

7. $\pi - \pi$

8. $\sqrt{\frac{81}{4}}$

5. _____

6. _____

7. _____

8. _____

Simplify each expression.

9. $31 - (5 - 3 \cdot 4)^2 + 2^6 \div 4$

10. $\frac{5 \cdot |-2 - 7(3)| - 2 \cdot 17}{22 + \sqrt{7} \cdot 28}$

9. _____

10. _____

11. _____

12. _____

11. $-a^2 + 4ab - b^2$ (if $a = 2$ and $b = -6$)

12. $2x - 5y$ (if $x = \frac{2}{3}$ and $y = \frac{1}{4}$)

Solve each equation.

13. $4x + 29 = 5 - 8(x + 6)$

14. $-\frac{4}{3}(9 - 2k) = \frac{1}{2}\left(\frac{1}{3}k + 1\right)$

13. _____

14. _____

15. _____

16. _____

15. $\frac{3m - 3}{5} = \frac{2m - 1}{2}$

16. $\frac{1}{3}a + b = 6a$ (solve for b)

Solve each equation. Be sure to check for extraneous solutions.

17. $|3 - 6w| = 39$

18. $-7 = \frac{|a - 3|}{-2}$

17. _____

18. _____

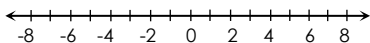
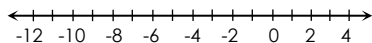
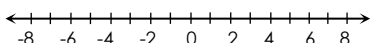
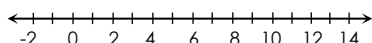
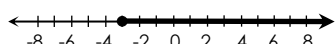
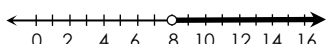
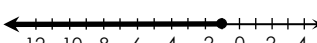
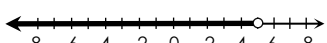
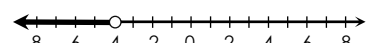
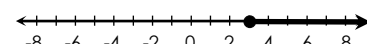
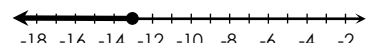
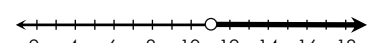
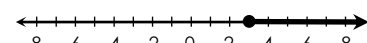
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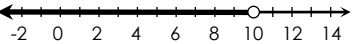
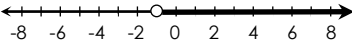
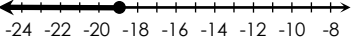
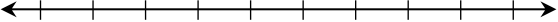
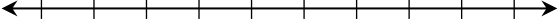
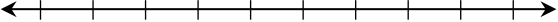
20. _____

19. $7|12x + 3| - 1 = 62$

20. $|5 - 3p| + 9 = 13p + 8$

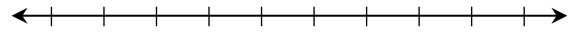
Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples		
Graphing INEQUALITIES	• Use a _____ dot for _____ or _____ symbols. • Use an _____ dot for _____ or _____ symbols.		
	Graph each inequality.		
	1. $x > 4$ 	2. $x \leq -7$ 	
	3. $x \geq -2$ 	4. $x < 11$ 	
INTERVAL & SET Notation	Interval Notation is a way to write the solution to an inequality using <u>infinity</u> symbols, <u>parentheses</u>, and <u>brackets</u>. • Parentheses indicate “not included” or “open” • Brackets indicate “included” or “closed”		
	Write each inequality in interval notation.		
	5. 	6. 	
	7. 	8. 	
	Set Notation is another way to write the solution to an inequality. Example: $x > 5$ Set Notation: _____ Read as _____ _____		
WRITING Inequalities	Write the inequality shown on the graph in interval and set notation.		
	Graph	Interval Notation	Set Notation
	1. 		
	2. 		
	3. 		
	4. 		
	5. 		

	6. 		
	7. 		
	8. 		
	Solve, graph, and write each solution in INTERVAL NOTATION.		
SOLVING <i>Multi-Step Inequalities</i> WATCH OUT: When you multiply or divide by a negative,	9. $5v + 1 > 7v - 17$	 Interval Notation:	
	10. $21x - 3(5x + 9) \geq 7$	 Interval Notation:	
	11. $12p + 52 > -2(4 - 2p)$	 Interval Notation:	
	12. $3 - (4m + 1) \geq 5m - 25$	 Interval Notation:	
	13. $3(8x - 9) > 4(6x + 1)$	 Interval Notation:	

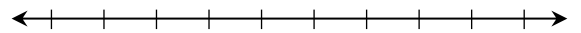
Solve, graph, and write each solution in SET NOTATION.

14. $10(a - 3) - 13a \leq 9(a - 8)$



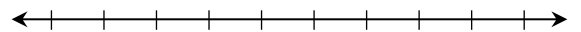
Set Notation:

15. $8 - 7(2r + 3) \leq 3(r - 10)$



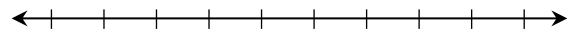
Set Notation:

16. $-3(2p + 8) < -4(2p + 6)$



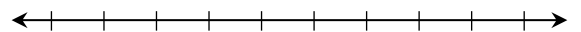
Set Notation:

17. $11x - 4 \leq 6x - (3 - 5x)$



Set Notation:

18. $\frac{4 + 5(2w - 10)}{-2} < w + 7$



Set Notation:

Name: _____

Unit 1: Equations & Inequalities

Date: _____ Per: _____

Homework 5: Multi-Step Inequalities

**** This is a 2-page document! ****

Solve, graph, and write the solution to each inequality in both INTERVAL and SET NOTATION.

1. $-7a - 11 < 1 - 4a$

Graph:

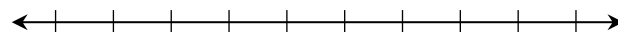


Interval Notation:

Set Notation:

2. $\frac{3}{4}(12v - 28) - v < -5$

Graph:

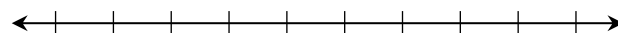


Interval Notation:

Set Notation:

3. $5 - 9(8 - 2x) \leq 2x - 115$

Graph:

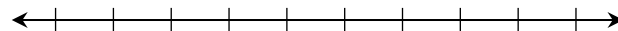


Interval Notation:

Set Notation:

4. $-2(7 + y) \geq -8(y + 1)$

Graph:

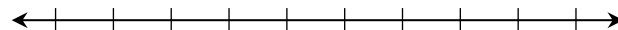


Interval Notation:

Set Notation:

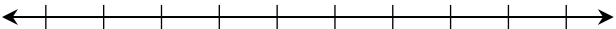
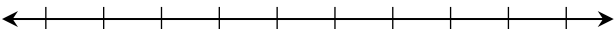
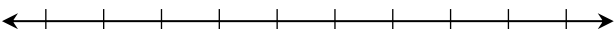
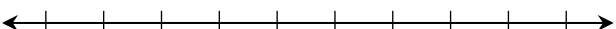
5. $-5(k + 4) > 2 - (3k + 6)$

Graph:

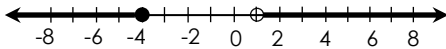
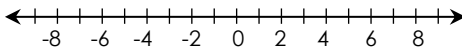
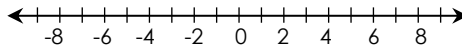
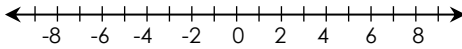
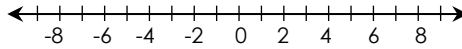
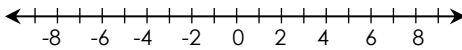
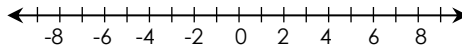


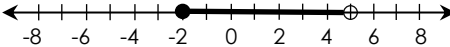
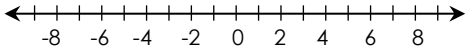
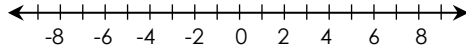
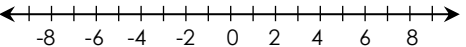
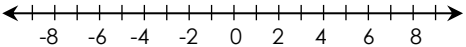
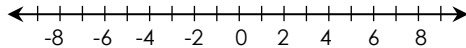
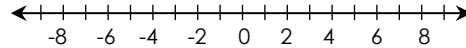
Interval Notation:

Set Notation:

6. $\frac{3(2x+7)}{-3} > x+1$	Graph:  <table border="1" data-bbox="870 373 1500 510"> <tr> <td>Interval Notation:</td><td>Set Notation:</td></tr> </table>	Interval Notation:	Set Notation:
Interval Notation:	Set Notation:		
7. $\frac{2}{3}p - 2(4-p) \geq 32$	Graph:  <table border="1" data-bbox="870 747 1500 884"> <tr> <td>Interval Notation:</td><td>Set Notation:</td></tr> </table>	Interval Notation:	Set Notation:
Interval Notation:	Set Notation:		
8. $r + 10 \leq -3(2r - 3) + 6(r + 3)$	Graph:  <table border="1" data-bbox="870 1121 1500 1257"> <tr> <td>Interval Notation:</td><td>Set Notation:</td></tr> </table>	Interval Notation:	Set Notation:
Interval Notation:	Set Notation:		
9. $-8m - 12 > -5(m - 8) - 3(m - 7)$	Graph:  <table border="1" data-bbox="870 1495 1500 1631"> <tr> <td>Interval Notation:</td><td>Set Notation:</td></tr> </table>	Interval Notation:	Set Notation:
Interval Notation:	Set Notation:		
10. $18\left(q - \frac{3}{4}\right) \leq -2\left(q + \frac{7}{4}\right)$	Graph:  <table border="1" data-bbox="870 1869 1500 2005"> <tr> <td>Interval Notation:</td><td>Set Notation:</td></tr> </table>	Interval Notation:	Set Notation:
Interval Notation:	Set Notation:		

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes	
COMPOUND <i>Inequalities</i>		
TYPE 1: “OR”		Inequality:
		Interval Notation:
		Set Notation:
SOLVING “OR” <i>Inequalities</i> To solve “or” inequalities, solve each part, then graph on a number line to show the solutions.	Solve, graph, and write each solution in interval notation.	
	1. $3x + 4 \leq 10$ or $5x > 45$	2. $n + 6 \leq 5$ or $9n + 7 > 43$
		
	Interval Notation:	Interval Notation:
	3. $-3y + 8 \leq -7$ or $10y - 3 \leq -33$	4. $6 - 2a < 8$ or $\frac{a}{2} + 1 < -2$
		
	Interval Notation:	Interval Notation:
	5. $3(5 - 2k) < -12$ or $\frac{1}{3}(15k - 9) \leq -3$	6. $\frac{8 - x}{-3} \leq -5$ or $5 - \frac{3}{2}x < -1$
		
Interval Notation:	Interval Notation:	

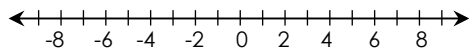
TYPE 2: “AND”		Inequality:
		Interval Notation:
		Set Notation:
REWRITING “AND” Inequalities	Because the solutions to an “and” inequality fall between two endpoints, they are frequently written in a more condensed form. Example: _____ ➡ _____	
SOLVING “AND” <i>Inequalities</i> If the “and” inequality is written out, solve each part separately. If condensed, you can solve it all together, working inside out. Then, graph to show all possible solutions.	Solve, graph, and write each solution in interval notation.	
	7. $2x + 3 > -9$ and $8x - 2 < 30$  Interval Notation:	8. $2w + 1 \leq 11$ and $1 - 5w < 41$  Interval Notation:
	9. $-2 \leq a - 1 \leq 6$  Interval Notation:	10. $4 < 3p + 4 < 19$  Interval Notation:
	11. $28 > -2 - 10m \geq -7$  Interval Notation:	12. $-5 \leq \frac{12 - x}{-2} < -2$  Interval Notation:

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes		
Special Cases of COMPOUND INEQUALITIES	We previously learned that compound inequalities are two inequalities joined by the word “or” or “and”. Now we will be looking at special cases of compound inequalities in which there is no solution, infinitely many solutions, or overlapping solutions.		
	Example at the Zoo: A sign at the ticket booth reads “Guests that are under 3 years old and over 62 years old get free admission!” Does this make sense? If not, why? How would you change it? Explain.		
	Graph the inequalities below and give your answer in interval notation.		
	Inequality	Graph	Interval Notation
	1. $x < -3$ and $x > 7$		
	2. $a \geq 0$ and $a < 6$		
	3. $k \leq 5$ and $k \leq 8$		
	4. $m > -4$ and $m \geq 2$		
	5. $r < -8$ or $r > 1$		
	6. $y \geq -2$ or $y > 7$		
	7. $n \leq 3$ or $n \leq -9$		
	8. $b > -1$ or $b < 1$		

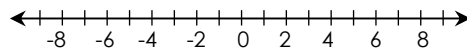
Solve, then graph the solutions and give your answer in interval notation.

9. $x - 4 > -1$ or $-8x < 48$



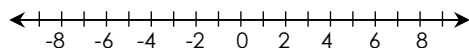
Interval Notation:

10. $\frac{p}{8} \leq -1$ and $p + 9 > 15$



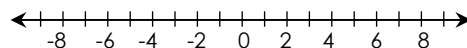
Interval Notation:

11. $7 + r < 10$ and $-2r - 1 \geq -3$



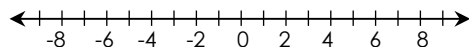
Interval Notation:

12. $\frac{a-4}{-2} \geq 1$ or $a - 1 \geq 6$



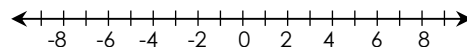
Interval Notation:

13. $5k + 3 > 8$ and $9k + 8 \geq -73$



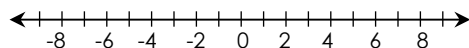
Interval Notation:

14. $1 - 2y \geq -17$ or $5y - 5 \leq -50$



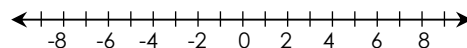
Interval Notation:

15. $2x + 6 > -4$ or $9x + 5 \leq 41$



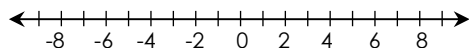
Interval Notation:

16. $10v + 4 < 44$ and $3v + 6 > -15$



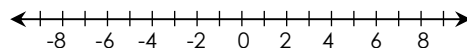
Interval Notation:

17. $10n + 1 \geq -59$ or $-10 - 9n < 26$



Interval Notation:

18. $6k - 4 \geq -46$ and $k + 5 \leq -5$



Interval Notation:

Name: _____

Unit 1: Equations & Inequalities

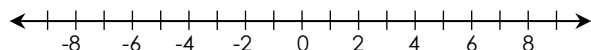
Date: _____ Per: _____

Homework 6: Compound Inequalities

**** This is a 2-page document! ****

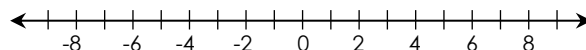
Solve, graph, and write the solution to each compound inequality in interval notation.

1. $-2x - 7 > 1$ or $x - 2 \geq -1$



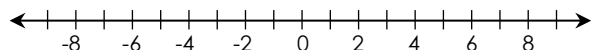
Interval Notation:

2. $\frac{a}{-2} < -1$ or $-4a + 3 \geq 23$



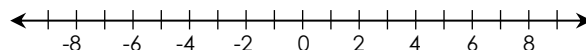
Interval Notation:

3. $6v + 38 \leq -4$ or $2(v + 3) \geq -2$



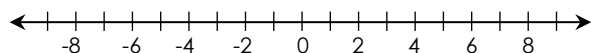
Interval Notation:

4. $4(1 - k) \geq -16$ or $7 - 6k < -41$



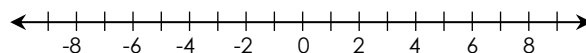
Interval Notation:

5. $10n - 9 > -59$ and $n - 6 \leq 3$



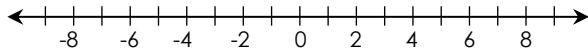
Interval Notation:

6. $-13 \leq 3 + 8p \leq 11$



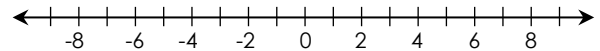
Interval Notation:

7. $37 > 7 - 6x > -17$



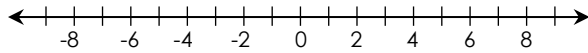
Interval Notation:

8. $3 \geq \frac{9-w}{3} > 1$



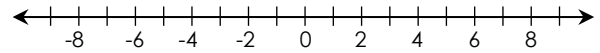
Interval Notation:

9. $2r + 6 \leq 4$ and $6r - 3 < 39$



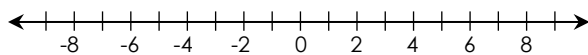
Interval Notation:

10. $6m + 8 > -10$ or $10m + 9 < 39$



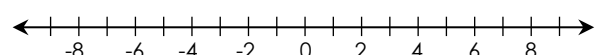
Interval Notation:

11. $6y + 2 < -4$ and $-4(2y + 3) < -68$



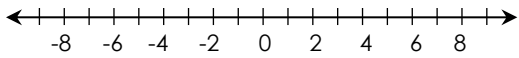
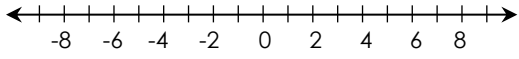
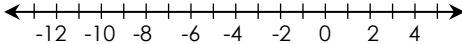
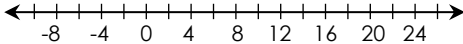
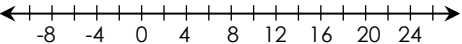
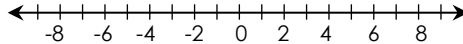
Interval Notation:

12. $\frac{4}{3}x + 11 \geq 7$ or $-4x - 3 \leq -23$

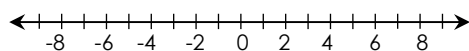


Interval Notation:

Name:	Date:
Topic:	Class:

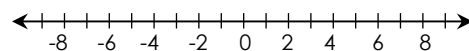
Main Ideas/Questions	Notes	
ABSOLUTE VALUE <i>Inequalities</i>	CASE 1	GREATER THAN/GREATER THAN OR EQUAL TO Example: $ x \geq 7$  Interval Notation:
		LESS THAN/LESS THAN OR EQUAL TO Example: $ x \leq 4$  Interval Notation:
	CASE 2	
What does this mean?		
SOLVING ABSOLUTE VALUE <i>Inequalities</i>	❶	ISOLATE the absolute value expression.
	❷	CREATE TWO CASES. Use the "KISS" Method (Keep it, switch, switch) to set up the two cases.
	❸	SOLVE both inequalities.
	❹	GRAPH your solutions and write your answer in interval notation.
Solve, graph, and write each solution in interval notation.		
1. $ x + 6 > 7$  Interval Notation:	2. $ 2v - 3 < 19$  Interval Notation:	
3. $\left -5 + \frac{k}{3} \right \geq 1$  Interval Notation:	4. $ 7 - 4n \leq 27$  Interval Notation:	

5. $\frac{|q-3|}{5} \geq 1$



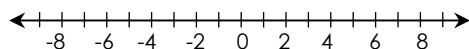
Interval Notation:

6. $|8a+10|+3 < 25$



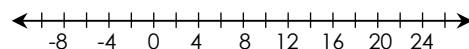
Interval Notation:

7. $-10|2r-1| \leq -60$



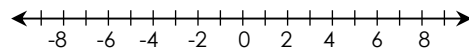
Interval Notation:

8. $9|3-k|-5 \geq 67$



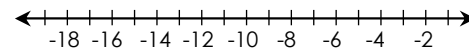
Interval Notation:

9. $6|-5n|+7 > 37$



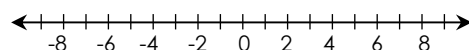
Interval Notation:

10. $-1-4|x+9| \leq -21$



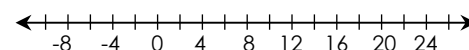
Interval Notation:

11. $3|8x+5|-9 \geq 24$



Interval Notation:

12. $4|9-3n|-9 < 75$



Interval Notation:

Name: _____

Unit 1: Equations & Inequalities

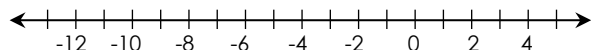
Date: _____ Per: _____

Homework 7: Absolute Value Inequalities

**** This is a 2-page document! ****

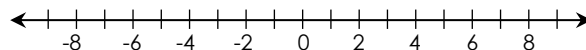
Solve, graph, and write the solution to each inequality in interval notation.

1. $|b + 3| \geq 7$



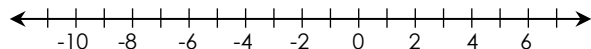
Interval Notation:

2. $|-2v - 4| < 8$



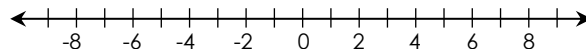
Interval Notation:

3. $|5k + 4| \geq 36$



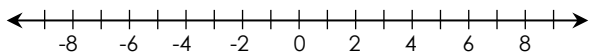
Interval Notation:

4. $|3 - 9y| \leq 33$



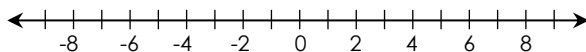
Interval Notation:

5. $\left|\frac{n}{7}\right| - 6 > -5$



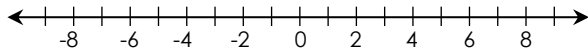
Interval Notation:

6. $\frac{|5 + 3w|}{-2} \leq -1$



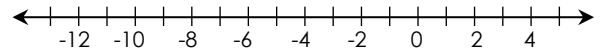
Interval Notation:

7. $7|4a - 6| < 42$



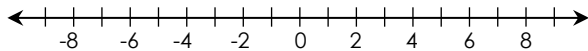
Interval Notation:

8. $2 - 9|p + 7| > -16$



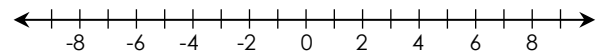
Interval Notation:

9. $3|z - 4| - 13 \leq -1$



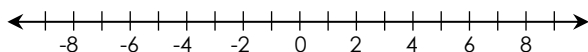
Interval Notation:

10. $\frac{|-2m|}{3} + 3 > 8$



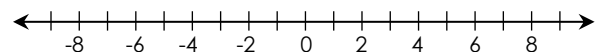
Interval Notation:

11. $4|8y + 7| - 9 < 91$



Interval Notation:

12. $2 - 5|10 - 2n| \geq -28$



Interval Notation:

Name: _____

Algebra 2

Date: _____ Per: _____

Unit 1: Equations & Inequalities

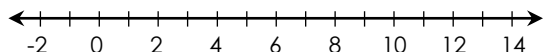
Quiz 1-2: Inequalities

Solve, graph, and write all final answers in interval notation.

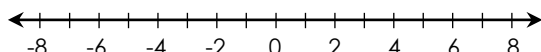
Part I: Multi-Step Inequalities

1. $5 - 4(2x - 7) < -39$

2. $11(m + 2) \leq 2m - (3m + 8)$



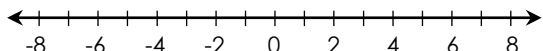
Interval Notation: _____



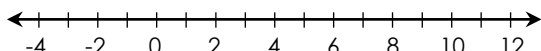
Interval Notation: _____

3. $2(7a - 1) - 11a \geq 3(a - 4)$

4. $\frac{5(8 - 2p) + 6}{-4} < p - 1$



Interval Notation: _____

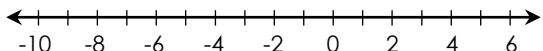


Interval Notation: _____

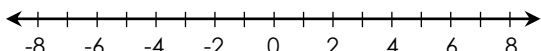
Part II: Compound Inequalities

5. $\frac{3}{4}x + 7 < 1$ or $9(4 - x) \leq 36$

6. $\frac{9 - 8w}{5} > 3$ or $\frac{2}{3}(18w - 3) < -62$



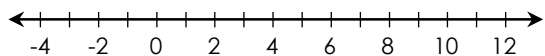
Interval Notation: _____



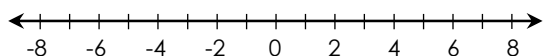
Interval Notation: _____

7. $7 - n \geq 6$ and $2(3n - 8) \geq 20$

8. $-5 < 10c - 1 < 74$



Interval Notation: _____

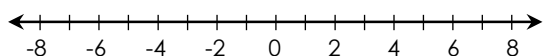


Interval Notation: _____

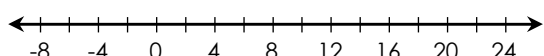
Part III: Absolute Value Inequalities

9. $|2k + 1| \geq 9$

10. $-7|x - 6| > -56$



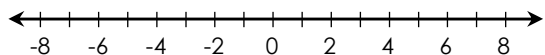
Interval Notation: _____



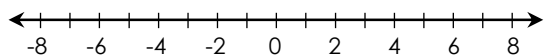
Interval Notation: _____

11. $|4m - 1| - 5 \leq 0$

12. $10 - 4|8y - 10| < -94$



Interval Notation: _____



Interval Notation: _____

Unit 1 Test Study Guide

(Equations & Inequalities)

Name: _____

Date: _____ Per: _____

Topic 1: Simplifying Radicals & Classifying Numbers in the Real Number System

Simplify each radical expression.			
1. $\sqrt{84}$	2. $\sqrt{243}$	3. $-\sqrt{160}$	4. $\sqrt{450}$
List ALL sets to which each value belongs. (Use R, I, Q, Z, W, N)			
5. $\frac{\sqrt{3}}{\sqrt{3}}$	6. $5.08\overline{3}$	7. $\frac{-\sqrt{256}}{2}$	8. 2π

Topic 2: Operations & Expressions

Simplify each expression.	
9. $[18 - (-1 - 7)^2] + 16 \div 2^4$	10. $\frac{4 + \sqrt{121} - 2 \cdot 3^3}{ -19 - 2(-8) }$
11. $10a - 5ab + 4b$ (if $a = \frac{2}{5}$ and $b = -\frac{1}{6}$)	12. $\sqrt{-x^2 - 4y^2}$ (if $x = 3$ and $y = -2$)

Topic 3: Multi-Step Equations & Word Problems

Solve each equation.	
13. $3(7 - 9k) + 23k = 4k - (24 - k)$	14. $7 - \frac{5}{2}(8n - 18) = 14 - 10(2n - 3)$

15. $\frac{6x-11}{7} = \frac{2x-9}{3}$	16. If $SA = \frac{1}{2}lp + B$, find p
17. The width of a rectangle is four less than one half its length. If the perimeter of the rectangle is 94 meters, find the area of the rectangle.	18. Find three consecutive odd numbers such that the sum of five times the smaller number and twice the larger number is 33 more than six times the median number.

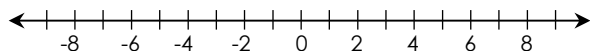
Topic 4: Absolute Value Equations

Solve each equation. Be sure to check for extraneous solutions.

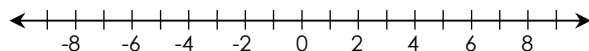
19. $ -7-4x = 17$	20. $\frac{ 5n-10 }{-2} = -15$
21. $2-10 k+1 = -78$	22. $ 2m+7 = 6m+13$

Topic 5: Multi-Step Inequalities**Solve and graph each inequality. Write your answer in interval notation.**

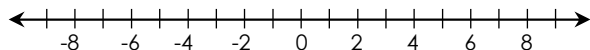
23. $33x - 8(3x + 9) > -9$

**Interval Notation:**

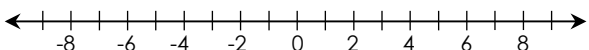
24. $-92 - 8k \geq 23 - (7 - 4k)$

**Interval Notation:**

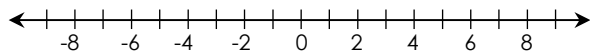
25. $11(5u - 4) - 7u \geq 8(6u - 7)$

**Interval Notation:**

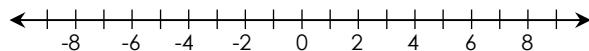
26. $-\frac{5}{3}\left(\frac{9}{10}x + 15\right) < 7 - \left(8 - \frac{9}{2}x\right)$

**Interval Notation:****Topic 6: Compound Inequalities****Solve and graph each compound inequality. Write your answer in interval notation.**

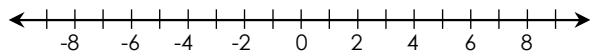
27. $-18 \leq 2b - 8 < -8$

**Interval Notation:**

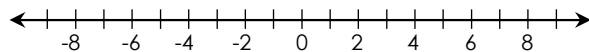
28. $m + 5 > 11$ or $8 - 10m \geq 33$

**Interval Notation:**

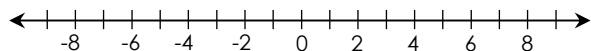
29. $7p + 5 \leq -37$ and $-10p < 10$

**Interval Notation:**

30. $7 - 3x \leq -20$ or $5x - 6 \leq 9$

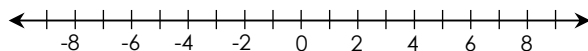
**Interval Notation:**

31. $9y - 2 < 34$ and $3y - 2 > -29$



Interval Notation:

32. $10 + 2w \geq 30$ or $5w - 8 > -13$

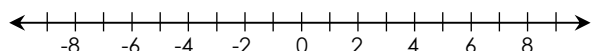


Interval Notation:

Topic 7: Absolute Value Inequalities

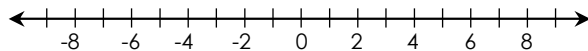
Solve and graph each absolute value inequality. Write your answer in interval notation.

33. $|7 - a| \geq 2$



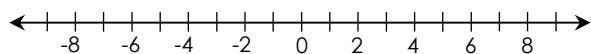
Interval Notation:

34. $\frac{|v - 3|}{-5} > -1$



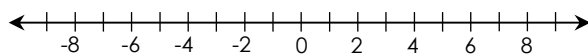
Interval Notation:

35. $|3n + 8| + 1 > 5$



Interval Notation:

36. $-3 - 6|4x - 10| < -87$



Interval Notation:

Name: _____

Date: _____ Per: _____

Unit 1 Test

Equations & Inequalities

1. Which value is a perfect square?

- A. 198
B. 249
C. 361
D. 500

2. If the expression below is written in simplest radical form, which must be true?

$$\sqrt{208} = \boxed{a} \sqrt{\boxed{b}}$$

- A. $a + b = 17$
B. $a + b = 23$
C. $a + b = 29$
D. $a + b = 31$

3. Which expression is correctly written in simplest form?

- A. $\sqrt{128} = 4\sqrt{8}$
B. $\sqrt{180} = 6\sqrt{5}$
C. $\sqrt{72} = 12\sqrt{3}$
D. $\sqrt{28} = 4\sqrt{7}$

4. Which set of numbers does **not** contain the value below?

$$-\frac{\sqrt{64}}{2}$$

- A. R
B. N
C. Z
D. Q

5. Which of the following lists of values contain rational numbers only?

- A. $90, -1.375, \pi$ C. $\frac{63}{9}, \sqrt{\frac{1}{16}}, \sqrt{50}$
B. $\sqrt{28}, 0, 3\frac{1}{5}$ D. $-\sqrt{81}, \frac{\sqrt{7}}{\sqrt{7}}, 4.\overline{16}$

6. Which is the smallest subset of the real numbers that contains the value below?

$$\sqrt{3} - \sqrt{3}$$

- A. R
B. N
C. W
D. Z

7. Simplify the expression below.

$$\sqrt{324} - (2 - 9)^2 + 162 \div 3^3$$

8. Simplify the expression below.

$$\frac{\sqrt{2^6} + (11 - 5^2)}{|-10 - 2(3)|}$$

9. Evaluate the expression below if $m = -4$, $n = 3$, and $p = 12$.

$$\frac{m^2n + 9p}{\sqrt{np}}$$

10. Evaluate the expression below if $x = \frac{4}{9}$, $y = \frac{1}{2}$, and $z = \frac{3}{5}$.

$$6xy - 3z$$

11. Solve the equation below.

$$8a - (1 + 5a) = 4(3a - 11) - 20$$

12. Solve the equation below.

$$8 - \frac{4}{3}\left(6 - \frac{5}{8}x\right) = \frac{1}{2}\left(\frac{7}{3}x + 18\right)$$

13. Which equation has a solution of all real numbers?

- A. $5k - (6k - 18) = 9 - k + 17$
B. $14 - 5(y + 10) = 4 - 8(y + 5)$
C. $6(-2w - 3) = -4(3w + 7) + 10$
D. $7m - 2(2m + 3) = 8 - (3m - 10)$

14. Solve the equation below.

$$\frac{5p - 8}{2} = \frac{7p + 4}{2}$$

15. Solve the equation below for m .

$$\frac{2m - n}{5} = 7n$$

A. $m = 40n$

B. $m = 18n$

C. $m = 8n$

D. $m = 20n$

16. If $2x - 1 = 13$, evaluate the expression below.

$$-x^2 - 4x$$

A. 12

B. 21

C. -60

D. -77

17. This past Sunday, the Giants scored 9 points less than twice the Cowboys. The Packers scored 14 more points than the Giants. If the teams scored 81 total points, find the number of points scored by the Packers.

18. The sum of three consecutive numbers is 41 less than five times the largest number. Find the largest number.

19. Solve the equation below.

$$|3x - 9| = 24$$

$x =$

20. Solve the equation below.

$$-2|7 - 9c| + 7 = -15$$

$c =$

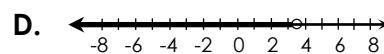
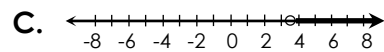
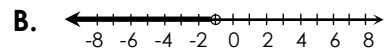
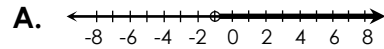
21. Solve the equation below.

$$|6w + 4| = 2w - 10$$

$w =$

22. Which graph represents the solution to the inequality below?

$$\frac{5 - (9 - 4x)}{-2} < -5$$



23. Which represents the solution to the inequality below in interval notation?

$$-\frac{1}{3}(21 - 9n) \geq 4(2n - 7) - 9$$

A. $(-\infty, 6]$

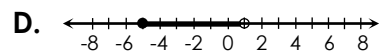
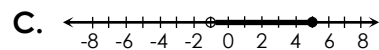
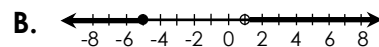
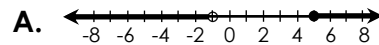
B. $(-\infty, -6]$

C. $[6, \infty)$

D. $[-6, \infty)$

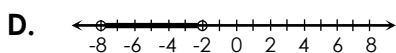
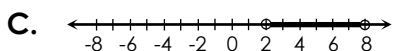
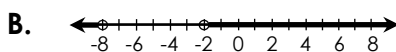
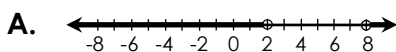
24. Which graph represents the solution to the compound inequality below?

$$-4k - 3 < -7 \text{ or } 1 - k \geq 6$$



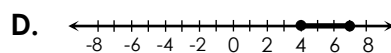
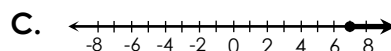
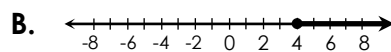
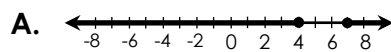
25. Which graph represents the solution to the compound inequality below?

$$-1 + 9y > -73 \text{ and } \frac{y - 4}{-3} > 2$$



26. Which graph represents the solution to the compound inequality below?

$$\frac{1}{2}a - 5 \geq -3 \text{ or } -6a - 14 \leq -56$$



27. Which represents the solution to the compound inequality below in interval notation?

$$5 - 2x > 31 \text{ and } 7x + 8 \geq -55$$

- A. $(-13, -9]$
- B. $(-\infty, -13) \cup [-9, \infty)$
- C. $[-9, \infty)$
- D. $\emptyset$

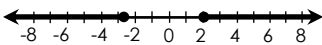
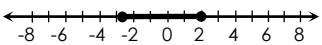
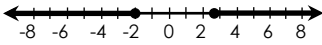
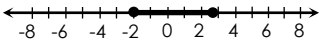
28. Which value of x is **not** in the solution to the compound inequality below?

$$-15 \leq 2x - 9 < 5$$

- A. -3
- B. 0
- C. 4
- D. 7

29. Which graph represents the solution to the inequality below?

$$|9x + 3| \leq 21$$

- A. 
- B. 
- C. 
- D. 

30. Which represents the solution to the inequality below in interval notation?

$$-3|n - 5| - 11 < -38$$

- A. $(-4, 14)$
- B. $(-14, 4)$
- C. $(-\infty, -4) \cup (14, \infty)$
- D. $(-\infty, -14) \cup (4, \infty)$

BONUS: A change jar contains nickels, dimes, and quarters, totaling an amount of \$1.90. The number of nickels is one more than twice the number of dimes. The number of quarters is one half the total number of nickels and dimes. Find the number of each coin in the change jar.

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes								
SQUARE ROOTS	The square root of a number, a, can be written as a radical expression: $\sqrt{\textcircled{a}}$ radical radicand								
Radicals with PERFECT SQUARES	List the first 20 perfect squares: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225, 256, 289, 324, 361, 400 Find each value. <table><tr><td>1. $\sqrt{25} = 5$</td><td>2. $\sqrt{256} = 16$</td><td>3. $-\sqrt{121} = -11$</td></tr><tr><td>4. $\sqrt{196} = 14$</td><td>5. $-\sqrt{324} = -18$</td><td>6. $\sqrt{\frac{9}{100}} = \frac{3}{10}$</td></tr></table>	1. $\sqrt{25} = 5$	2. $\sqrt{256} = 16$	3. $-\sqrt{121} = -11$	4. $\sqrt{196} = 14$	5. $-\sqrt{324} = -18$	6. $\sqrt{\frac{9}{100}} = \frac{3}{10}$		
1. $\sqrt{25} = 5$	2. $\sqrt{256} = 16$	3. $-\sqrt{121} = -11$							
4. $\sqrt{196} = 14$	5. $-\sqrt{324} = -18$	6. $\sqrt{\frac{9}{100}} = \frac{3}{10}$							
SIMPLIFYING Radicals	If the radicand is NOT a perfect square, we can simplify the radical. 1 Find the largest perfect square that the radicand is divisible by. Break down the radical using this number. 2 Take the square root of the perfect square. Take it out of the radical. 3 Leave the "leftover" under the radical symbol. Simplify each expression <table><tr><td>7. $\sqrt{44} = \sqrt{4} \cdot \sqrt{11}$ $= 2\sqrt{11}$</td><td>8. $\sqrt{288} = \sqrt{144} \cdot \sqrt{2}$ $= 12\sqrt{2}$</td></tr><tr><td>9. $\sqrt{147} = \sqrt{49} \cdot \sqrt{3}$ $= 7\sqrt{3}$</td><td>10. $-\sqrt{80} = -\sqrt{16} \cdot \sqrt{5}$ $= -4\sqrt{5}$</td></tr><tr><td>11. $\sqrt{600} = \sqrt{100} \cdot \sqrt{6}$ $= 10\sqrt{6}$</td><td>12. $\sqrt{162} = \sqrt{81} \cdot \sqrt{2}$ $= 9\sqrt{2}$</td></tr><tr><td>13. $-\sqrt{75} = -\sqrt{25} \cdot \sqrt{3}$ $= -5\sqrt{3}$</td><td>14. $\sqrt{448} = \sqrt{64} \cdot \sqrt{7}$ $= 8\sqrt{7}$</td></tr></table>	7. $\sqrt{44} = \sqrt{4} \cdot \sqrt{11}$ $= 2\sqrt{11}$	8. $\sqrt{288} = \sqrt{144} \cdot \sqrt{2}$ $= 12\sqrt{2}$	9. $\sqrt{147} = \sqrt{49} \cdot \sqrt{3}$ $= 7\sqrt{3}$	10. $-\sqrt{80} = -\sqrt{16} \cdot \sqrt{5}$ $= -4\sqrt{5}$	11. $\sqrt{600} = \sqrt{100} \cdot \sqrt{6}$ $= 10\sqrt{6}$	12. $\sqrt{162} = \sqrt{81} \cdot \sqrt{2}$ $= 9\sqrt{2}$	13. $-\sqrt{75} = -\sqrt{25} \cdot \sqrt{3}$ $= -5\sqrt{3}$	14. $\sqrt{448} = \sqrt{64} \cdot \sqrt{7}$ $= 8\sqrt{7}$
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REAL NUMBERS

(R)

The **Real Numbers** consist of the following sets of numbers:

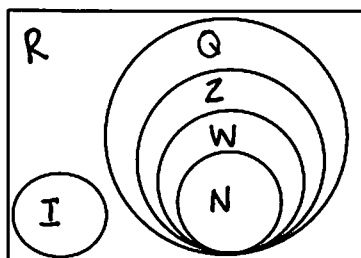
- **Irrational Numbers:** Numbers in which the decimal form never terminates and does not repeat. Common examples are π and non-perfect square roots.
- **Rational Numbers:** Numbers in which the decimal form either terminates or repeats. Rational numbers can always be written as a **fraction** (a/b) where a and b are integers.

Subsets of Rational Numbers:

- **Integers:** $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$
- **Whole Numbers:** $\{0, 1, 2, 3, 4, \dots\}$
- **Natural Numbers:** $\{1, 2, 3, 4, 5, \dots\}$

ORGANIZING

the Real Numbers



Set	Letter	Examples
Irrational Numbers	I	$\sqrt{2}, \pi, \sqrt{72}, \sqrt{41}$
Rational Numbers	Q	$0.34, 1\frac{5}{8}, 0.\bar{2}, -\frac{7}{17}$
Integers	Z	$7, -\sqrt{100}, 0, \frac{8}{4}$
Whole Numbers	W	$0, \frac{20}{4}, \sqrt{16}, 90$
Natural Numbers	N	$10, 47, \sqrt{9}, \frac{34}{2}$

CLASSIFYING

Numbers

Name all sets to which each value belongs.

15. -10 R, Q, Z	16. $\sqrt{30}$ R, I	17. $1\frac{7}{9}$ R, Q
18. $\frac{24}{8}$ R, Q, Z, W, N	19. 2.48 R, Q	20. $-\sqrt{81} = -9$ R, Q, Z
21. $\pi - \pi = 0$ R, Q, Z, W	22. $\sqrt{\frac{16}{25}} = \frac{4}{5}$ R, Q	23. $0.\bar{15}$ R, Q
24. 109 R, Q, Z, W, N	25. $-\frac{8}{3}$ R, Q	26. $\frac{\sqrt{7}}{\sqrt{7}} = 1$ R, Q, Z, W, N

Name: _____

Unit 1: Equations & Inequalities

Date: _____ Per: _____

Homework 1: Simplifying Radicals & Classifying Numbers

Simplify each expression.		
1. $\sqrt{64} = 8$	2. $-\sqrt{225} = -15$	3. $\sqrt{400} = 20$
4. $\sqrt{27} = \sqrt{9} \cdot \sqrt{3}$ $= 3\sqrt{3}$	5. $\sqrt{363} = \sqrt{121} \cdot \sqrt{3}$ $= 11\sqrt{3}$	6. $-\sqrt{216} = -\sqrt{36} \cdot \sqrt{6}$ $= -6\sqrt{6}$
7. $\sqrt{125} = \sqrt{25} \cdot \sqrt{5}$ $= 5\sqrt{5}$	8. $-\sqrt{32} = -\sqrt{16} \cdot \sqrt{2}$ $= -4\sqrt{2}$	9. $\sqrt{245} = \sqrt{49} \cdot \sqrt{5}$ $= 7\sqrt{5}$
10. $\sqrt{96} = \sqrt{16} \cdot \sqrt{6}$ $= 4\sqrt{6}$	11. $\sqrt{392} = \sqrt{196} \cdot \sqrt{2}$ $= 14\sqrt{2}$	12. $-\sqrt{192} = -\sqrt{64} \cdot \sqrt{3}$ $= -8\sqrt{3}$

Name ALL SETS to which each number belongs.

13. $-\frac{5}{3}$ R, Q	14. $\sqrt{49}$ R, Q, Z, W, N	15. $0.\bar{6}$ R, Q
16. π R, I	17. $-\frac{36}{4} = -9$ R, Q, Z	18. $\sqrt{90}$ R, I
19. 1.125 R, Q	20. $-\sqrt{169} = -13$ R, Q, Z	21. $\frac{4}{11}$ R, Q

22. Place the LETTER of each value on the diagram in the smallest subset that contains the value.

A. $-\sqrt{196} = -14$ B. $\frac{\pi}{\pi} = 1$ C. $2\frac{6}{7}$ D. $\sqrt{17}$ E. $1.8\bar{3}$ F. -0.75 G. $\sqrt{2} - \sqrt{2} = 0$ H. $\sqrt{\frac{1}{4}} = \frac{1}{2}$ I. $\frac{35}{7} = 5$ J. $\frac{1}{13}$	
--	--

Name:

Date:

Topic:

Class:

Main Ideas/Questions

Notes

Order of OPERATIONS

G	Grouping Symbols (), [], { }, and fraction bars
E	Exponents (and Roots)
MD	Multiplication & Division (left → right)
AS	Addition & Subtraction (left → right)

Simplify each expression.

$$\begin{aligned}
 1. \quad & 29 - (11 - 2^3) + 6^2 \div 4 \\
 & = 29 - (11 - 8) + 36 \div 4 \\
 & = 29 - 3 + 9 \\
 & = 26 + 9 \\
 & = \boxed{35}
 \end{aligned}$$

$$\begin{aligned}
 2. \quad & 4^3 - |-8| \cdot 6 \div 16 \\
 & = 64 - 8 \cdot 6 \div 16 \\
 & = 64 - 48 \div 16 \\
 & = 64 - 3 \\
 & = \boxed{61}
 \end{aligned}$$

$$\begin{aligned}
 3. \quad & \frac{7 - (|10 - 12| + 5^2)}{9 - 2^2} \\
 & = \frac{7 - (2 + 25)}{9 - 4} \\
 & = \frac{7 - 27}{5} \\
 & = \frac{-20}{5} = \boxed{-4}
 \end{aligned}$$

$$\begin{aligned}
 4. \quad & \frac{(2 - 3^2)^2 + \sqrt{81}}{10 - 6 \div 3} \\
 & = \frac{(2 - 9)^2 + 9}{10 - 2} \\
 & = \frac{49 + 9}{8} \\
 & = \frac{58}{8} = \boxed{\frac{29}{4}}
 \end{aligned}$$

$$\begin{aligned}
 5. \quad & \frac{[15 - (|-6 + 2| + \sqrt{9})]^2}{3 - 3^3} \\
 & = \frac{[15 - (4 + 3)]^2}{3 - 27} \\
 & = \frac{[15 - 7]^2}{-24} = \frac{64}{-24} = \boxed{-\frac{8}{3}}
 \end{aligned}$$

$$\begin{aligned}
 6. \quad & \frac{(9 - 15)^2 \cdot (7 - 2^2)}{\sqrt{49} + 5} - |-5| \\
 & = \frac{(-6)^2 \cdot (7 - 4)}{7 + 5} - 5 \\
 & = \frac{36 \cdot 3}{12} - 5 \\
 & = \frac{108}{12} - 5 = 9 - 5 = \boxed{4}
 \end{aligned}$$

EVALUATING Expressions

Evaluate each expression given the variable replacements.

$$\begin{aligned}
 7. \quad & \frac{a \cdot b^2}{c + 12} \quad (\text{if } a = 3, b = -4, \text{ and } c = 8) \\
 & = \frac{3(-4)^2}{8 + 12} \\
 & = \frac{3(16)}{20} = \frac{48}{20} = \boxed{\frac{12}{5}}
 \end{aligned}$$

$$\begin{aligned}
 8. \quad & |10m - 8n^2| \quad (\text{if } m = 2 \text{ and } n = -3) \\
 & = |10(2) - 8(-3)^2| \\
 & = |20 - 8(9)| \\
 & = |20 - 72| \\
 & = |-52| = \boxed{52}
 \end{aligned}$$

	9. $-x^2 + 7(y+2) - 1$ (if $x = 5$ and $y = -9$) $= -(5)^2 + 7(-9+2) - 1$ $= -25 + 7(-7) - 1$ $= -25 - 49 - 1$ $= -74 - 1$ $= \boxed{-75}$	10. $3r - 4s$ (if $r = 5$ and $s = \frac{2}{3}$) $= 3(5) - 4(\frac{2}{3})$ $= 15 - \frac{8}{3}$ $= \frac{45}{3} - \frac{8}{3}$ $= \boxed{\frac{37}{3}}$
	11. $x^2 - y^2 - x \cdot y$ (if $x = -6$ and $y = 8$) $= (-6)^2 - (8)^2 - (-6)(8)$ $= 36 - 64 - (-48)$ $= -28 + 48$ $= 28 + 48 = \boxed{76}$	12. $\frac{2}{5}(a+2b)^2$ (if $a = 3$ and $b = -4$) $= \frac{2}{5}(3 + 2(-4))^2$ $= \frac{2}{5}(3 - 8)^2$ $= \frac{2}{5}(25)$ $= \boxed{10}$
	13. $\frac{w^3 - 4v}{\sqrt{wv}}$ (if $w = 2$ and $v = 18$) $= \frac{(2)^3 - 4(18)}{\sqrt{2 \cdot 18}}$ $= \frac{8 - 72}{\sqrt{36}}$ $= \frac{-64}{6} = \boxed{-\frac{32}{3}}$	14. $2m^2 + 9m$ (if $m = \frac{1}{4}$) $2(\frac{1}{4})^2 + 9(\frac{1}{4})$ $= 2(\frac{1}{16}) + \frac{9}{4}$ $= \frac{1}{8} + \frac{9}{4}$ $= \frac{1}{8} + \frac{18}{8} = \boxed{\frac{19}{8}}$
	Simplify each expression by distributing and/or combining like terms.	
SIMPLIFYING Expressions	15. $16x - 11 - 7x + 10$ $= \boxed{9x - 1}$	16. $13 + 2(3n - 7) + 19n$ $= 13 + 6n - 14 + 19n$ $= \boxed{25n - 1}$
	17. $6(c - 2) - (8c + 1)$ $= 6c - 12 - 8c - 1$ $= \boxed{-2c - 13}$	18. $-15y - 7z + y + 9z$ $= \boxed{-14y + 2z}$
	19. $3(3a - 5b) - (a - 7b)$ $= 9a - 15b - a + 7b$ $= \boxed{8a - 8b}$	20. $-8(2x + y) - 3(-5x - 2y)$ $= -16x - 8y + 15x + 6y$ $= \boxed{-x - 2y}$

Name: _____

Unit 1: Equations & Inequalities

Date: _____ Per: _____

Homework 2: Expressions & Operations

**** This is a 2-page document! ******Simplify each expression.**

$$\begin{aligned}
 1. & 25 - (\sqrt{16} - 1) \cdot (3 - 9)^2 \\
 & = 25 - (4 - 1) \cdot (-6)^2 \\
 & = 25 - 3 \cdot 36 \\
 & = 25 - 108 \\
 & = \boxed{-83}
 \end{aligned}$$

$$\begin{aligned}
 2. & |5 - 2(9)| + 9^2 \div 3 \\
 & = |5 - 18| + 81 \div 3 \\
 & = 13 + 27 \\
 & = \boxed{40}
 \end{aligned}$$

$$\begin{aligned}
 3. & \frac{2^5 - 4 \cdot 3^3}{7 + (1 - \sqrt{100})} \\
 & = \frac{32 - 4 \cdot 27}{7 + (1 - 10)} \\
 & = \frac{32 - 108}{7 - 9} \\
 & = \frac{-76}{-2} = \boxed{38}
 \end{aligned}$$

$$\begin{aligned}
 4. & \frac{(\sqrt{225} - 11) \cdot 12}{-12 - (8 - 6^2)} - |-3| \\
 & = \frac{(15 - 11) \cdot 12}{-12 - (8 - 36)} - 3 \\
 & = \frac{4 \cdot 12}{-12 - (-28)} - 3 \\
 & = \frac{48}{16} - 3 = \boxed{0}
 \end{aligned}$$

$$\begin{aligned}
 5. & \frac{3}{2} [(58 - 10^2) \div (\sqrt{16} + 3)] \\
 & = \frac{3}{2} [(58 - 100) \div (4 + 3)] \\
 & = \frac{3}{2} [-42 \div 7] \\
 & = \frac{3}{2} [-6] = \boxed{-9}
 \end{aligned}$$

$$\begin{aligned}
 6. & \frac{56 \div (7 - 9)^3 - 25}{24 - 5 \cdot 4} \\
 & = \frac{56 \div (-2)^3 - 25}{24 - 20} \\
 & = \frac{56 \div (-8) - 25}{4} \\
 & = \frac{-7 - 25}{4} = \frac{-32}{4} = \boxed{-8}
 \end{aligned}$$

Evaluate each expression given the replacement values.

$$\begin{aligned}
 7. & a^2 + 2(b - 6) - 17 \quad (\text{if } a = -7 \text{ and } b = 2) \\
 & = (-7)^2 + 2(2 - 6) - 17 \\
 & = 49 + 2(-4) - 17 \\
 & = 49 - 8 - 17 \\
 & = 41 - 17 = \boxed{24}
 \end{aligned}$$

$$\begin{aligned}
 8. & \frac{10xy}{8x - 2y + 10} \quad (\text{if } x = 4 \text{ and } y = -7) \\
 & = \frac{10(4)(-7)}{8(4) - 2(-7) + 10} \\
 & = \frac{-280}{32 + 14 + 10} \\
 & = \frac{-280}{56} = \boxed{-5}
 \end{aligned}$$

9. $\frac{3}{4} m - 2n^2$ (if $m = -4$ and $n = 6$) $= \frac{3}{4} -4 - 2(6)^2 $ $= \frac{3}{4} -4 - 72 $ $= \frac{3}{4} (76) = \boxed{57}$	10. $\frac{(4c + 9d)^2}{c^2 - 3d^2}$ (if $c = 3$ and $d = -2$) $= \frac{(4(3) + 9(-2))^2}{(3)^2 - 3(-2)^2}$ $= \frac{(12 - 18)^2}{9 - 12} = \frac{(-6)^2}{-3} = \frac{36}{-3} = \boxed{-12}$
11. $-w^2 - 11uv + 2v^2$ (if $u = -1$, $v = 8$, and $w = -3$) $= -(-3)^2 - 11(-1)(8) + 2(8)^2$ $= -9 + 88 + 2(64)$ $= 79 + 128$ $= \boxed{207}$	12. $\frac{2}{5}(x - 4y) + \frac{1}{2}(2x + y)$ (if $x = 1$ and $y = -6$) $= \frac{2}{5}(1 - 4(-6)) + \frac{1}{2}(2(1) + (-6))$ $= \frac{2}{5}(1 + 24) + \frac{1}{2}(2 - 6)$ $= \frac{2}{5}(25) + \frac{1}{2}(-4)$ $= 10 - 2 = \boxed{8}$
13. $\frac{a^3 - \sqrt{2ab}}{(a - b)^2}$ (if $a = 4$ and $b = 8$) $= \frac{(4)^3 - \sqrt{2(4)(8)}}{(4 - 8)^2}$ $= \frac{64 - \sqrt{64}}{(-4)^2} = \frac{64 - 8}{16} = \frac{56}{16} = \boxed{\frac{7}{2}}$	14. $p^2 + 2pq - 3q^2$ (if $p = \frac{1}{4}$ and $q = \frac{2}{3}$) $= \left(\frac{1}{4}\right)^2 + 2\left(\frac{1}{4}\right)\left(\frac{2}{3}\right) - 3\left(\frac{2}{3}\right)^2$ $= \frac{1}{16} + \frac{4}{12} - \frac{12}{9}$ $= \frac{1}{16} + \frac{1}{3} - \frac{4}{3} = \boxed{\frac{-15}{16}}$
Simplify each expression below by distributing and/or combining like terms.	
15. $9x - 2 + 3x - 5$ $= \boxed{12x - 7}$	16. $-2(6m - 7) + 3(m - 13)$ $= -12m + 14 + 3m - 39$ $= \boxed{-9m - 25}$
17. $17 - (2w + 3) + 7(10 - w)$ $= 17 - 2w - 3 + 70 - 7w$ $= \boxed{-9w + 84}$	18. $19x + 2y - 10y - 7x$ $= \boxed{12x - 8y}$
19. $3(-2a + 8b) - (9b - a)$ $= -6a + 24b - 9b + a$ $= \boxed{-5a + 15b}$	20. $7x(5 - y) - 3(9x + 2) - 4xy$ $= 35x - 7xy - 27x - 6 - 4xy$ $= \boxed{8x - 11xy - 6}$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Examples											
Solving EQUATIONS <table><tr><th colspan="2">Basic Steps to Follow</th></tr><tr><td>1</td><td>Distribute</td></tr><tr><td>2</td><td>Simplify Each Side by combining like terms.</td></tr><tr><td>3</td><td>Move Variables to one side.</td></tr><tr><td>4</td><td>Solve using inverse operations.</td></tr></table>	Basic Steps to Follow		1	Distribute	2	Simplify Each Side by combining like terms.	3	Move Variables to one side.	4	Solve using inverse operations.	Directions: Solve each equation below.	
	Basic Steps to Follow											
	1	Distribute										
	2	Simplify Each Side by combining like terms.										
	3	Move Variables to one side.										
4	Solve using inverse operations.											
	1. $6(n + 5) = -18$ $6n + 30 = -18$ $6n = -48$ $n = -8$	2. $\frac{3(5 - 3x)}{-4} = -15$ $\frac{15 - 9x}{-4} = -15$ $15 - 9x = 60$ $-9x = 45$ $x = -5$										
	3. $2(2 - 7y) - 5 = -29$ $4 - 14y - 5 = -29$ $-14y - 1 = -29$ $-14y = -28$ $y = 2$	4. $-16 = -\frac{15}{16}\left(\frac{4}{5}q + \frac{32}{3}\right)$ $-16 = -\frac{3}{4}q - 10$ $-6 = -\frac{3}{4}q$ $-24 = -3q$ $8 = q$										
	5. $10a - 7 = 21 - 6a$ $16a - 7 = 21$ $16a = 28$ $a = \frac{7}{4}$	6. $\frac{5}{6}w + 21 = -\frac{1}{3}(2w - 9)$ $\left[\frac{5}{6}w + 21 = -\frac{2}{3}w + 3\right] \cdot 6$ $5w + 126 = -4w + 18$ $9w + 126 = 18$ $9w = -108$ $w = -12$										
	7. $-5(8p + 6) = 5 - 5p$ $-40p - 30 = 5 - 5p$ $-35p - 30 = 5$ $-35p = 35$ $p = -1$	8. $16 - (7m + 3) = 8(1 - m)$ $16 - 7m - 3 = 8 - 8m$ $-7m + 13 = 8 - 8m$ $m + 13 = 8$ $m = -5$										

	9. $\frac{10x+17-(x-9)}{2} = 3x+1$ $10x+17-x+9 = 6x+2$ $9x+26 = 6x+2$ $3x+26 = 2$ $3x = -24$ $\boxed{x = -8}$	10. $\frac{9}{8}\left(4 - \frac{1}{2}a\right) = 3\left(\frac{1}{4}a + \frac{5}{2}\right)$ $\left[\frac{9}{2} - \frac{9}{16}a = \frac{3}{4}a + \frac{15}{2}\right] \cdot 16$ $72 - 9a = 12a + 120$ $72 = 21a + 120$ $-48 = 21a$ $\boxed{-\frac{16}{7} = a}$
SPECIAL Cases	11. $1 + 3c = 5(1 - c) + 8c$ $1 + 3c = 5 - 5c + 8c$ $1 + 3c = 5 + 3c$ $1 \neq 5$ $\boxed{\emptyset; \text{No Solution}}$	12. $40 - 8x = -8(x - 5)$ $40 - 8x = -8x + 40$ $40 = 40$ $\boxed{\mathbb{R}; \text{all real \#s}}$
PROPORTIONS	13. $\frac{5}{2} = \frac{40}{x+4}$ $5(x+4) = 2(40)$ $5x + 20 = 80$ $5x = 60$ $\boxed{x = 12}$	14. $\frac{10}{5x-9} = \frac{8}{2x-8}$ $10(2x-8) = 8(5x-9)$ $20x - 80 = 40x - 72$ $-80 = 20x - 72$ $-8 = 20x$ $\boxed{-\frac{2}{5} = x}$
LITERAL Equations	Directions: Solve each equation or formula for the specified variable. 15. $\frac{C}{(2r)} = \frac{2\pi r}{(2r)}$ (for π) $\boxed{\frac{C}{2r} = \pi}$	16. $A = \frac{1}{2}ap$ (for p) $2A = ap$ $\boxed{\frac{2A}{a} = p}$
	17. $\frac{a+b}{c} = 5$ (for b) $a+b = 5c$ $\boxed{b = 5c - a}$	18. $3x - 3y = 24$ (for x) $\frac{3x}{3} = \frac{24 + 3y}{3}$ $\boxed{x = 8 + y}$
	19. $7w = \frac{1}{2}(w+v)$ (for v) $14w = w + v$ $\boxed{13w = v}$	20. $S = (n-2) \cdot 180$ (for n) $\frac{S}{180} = n - 2$ $\boxed{\frac{S}{180} + 2 = n}$

USING EQUATIONS to solve WORD PROBLEMS

Directions: Set up and solve an equation for each problem below.

1. The product of 7 and a number, increased by 3, is equal to twice the number, subtracted from 39. Find the number.

let $x = \text{number}$

$$7x + 3 = 39 - 2x$$

$$9x + 3 = 39$$

$$9x = 36$$

$$\boxed{x = 4}$$

2. Five less than twice a number is equal to one half the difference of three times the number and 13. Find the number.

let $x = \text{number}$

$$2x - 5 = \frac{1}{2}(3x - 13)$$

$$4x - 10 = 3x - 13$$

$$x - 10 = -13$$

$$\boxed{x = -3}$$

3. One number is 9 more than twice another number. If the sum of the numbers is 129, find both numbers.

let $x = \text{number}$

let $2x + 9 = \text{other number}$

$$x + 2x + 9 = 129$$

$$3x + 9 = 129$$

$$3x = 120$$

$$x = 40$$

$$\boxed{40, 89}$$

4. Kara's golf score was 1 less than three fourths of Greg's golf score. If the sum of their scores was 153, find Kara's golf score.

let $x = \text{Greg's score}$

let $\frac{3}{4}x - 1 = \text{Kara's score}$

$$x + \frac{3}{4}x - 1 = 153$$

$$\frac{7}{4}x - 1 = 153$$

$$\frac{7}{4}x = 154$$

$$7x = 616$$

$$x = 88$$

$$\boxed{\text{Kara} = 65}$$

5. The length of a rectangle is one more than four times its width. If the perimeter of the rectangle is 62 meters, find the dimensions of the rectangle.

let $x = \text{width}$

let $4x + 1 = \text{length}$

$$2(x) + 2(4x + 1) = 62$$

$$2x + 8x + 2 = 62$$

$$10x + 2 = 62$$

$$10x = 60$$

$$x = 6$$

$$\boxed{\begin{array}{l} \text{width} = 6\text{m} \\ \text{length} = 25\text{m} \end{array}}$$

6. The length of one base of a trapezoid is 19 less than five times the length of the other base. If the trapezoid has a height of 18 feet and an area of 477 ft², find the length of the longer base.

let $x = \text{base 1}$

let $5x - 19 = \text{base 2}$

$$477 = \frac{1}{2}(x + 5x - 19) \cdot 18$$

$$477 = \frac{1}{2} \cdot 18 \cdot (6x - 19)$$

$$477 = 9(6x - 19)$$

$$477 = 54x - 171$$

$$648 = 54x$$

$$x = 12$$

$$\boxed{\begin{array}{l} \text{longer base} \\ = 41 \text{ ft} \end{array}}$$

7. The measure of one angle is six more than half the measure of another angle. If the angles are complementary, find both measures.

let $x = \text{angle}$

let $\frac{1}{2}x + 6 = \text{other angle}$

$$x + \frac{1}{2}x + 6 = 90$$

$$\frac{3}{2}x + 6 = 90$$

$$\frac{3}{2}x = 84$$

$$3x = 168$$

$$x = 56$$

$$\boxed{56^\circ, 34^\circ}$$

8. Kelly, Laura, and Tia sold boxes of cookies for a fundraiser. Kelly sold 15 boxes less than Laura. Tia sold twice as many boxes as Kelly. If they sold 83 boxes of cookies in total, how many boxes did each girl sell?

let $x = \text{Laura}$

let $x - 15 = \text{Kelly}$

let $2x - 30 = \text{Tia}$

$$x + x - 15 + 2x - 30 = 83$$

$$4x - 45 = 83$$

$$4x = 128$$

$$x = 32$$

$$\boxed{\text{Laura} = 32, \text{Kelly} = 17, \text{Tia} = 34}$$

9. The measure of the vertex angle in an isosceles triangle is six less than four times the measure of a base angle. Find the measure of the vertex angle.

let $x = \text{base angle}$

let $4x - 6 = \text{vertex angle}$

$$2(x) + 4x - 6 = 180$$

$$6x - 6 = 186$$

$$6x = 192$$

$$x = 32$$

$$\boxed{\text{Vertex angle} = 118^\circ}$$

10. The sum of four consecutive integers is 222. Find the integers.

let $x = 1^{\text{st}} \text{ integer}$

let $x + 1 = 2^{\text{nd}} \text{ integer}$

let $x + 2 = 3^{\text{rd}} \text{ integer}$

let $x + 3 = 4^{\text{th}} \text{ integer}$

$$x + x + 1 + x + 2 + x + 3 = 222$$

$$4x + 6 = 222$$

$$4x = 216$$

$$x = 54$$

$$\boxed{54, 55, 56, 57}$$

11. The sum of three consecutive odd integers is 93. Find the integers.

let $x = 1^{\text{st}} \text{ odd integer}$

let $x + 2 = 2^{\text{nd}} \text{ odd integer}$

let $x + 4 = 3^{\text{rd}} \text{ odd integer}$

$$x + x + 2 + x + 4 = 93$$

$$3x + 6 = 93$$

$$3x = 87$$

$$x = 29$$

$$\boxed{29, 31, 33}$$

12. Find three consecutive even integers such that the sum of the smallest integer and twice the median integer is 20 more than the largest integer.

let $x = 1^{\text{st}} \text{ even integer}$

let $x + 2 = 2^{\text{nd}} \text{ even integer}$

let $x + 4 = 3^{\text{rd}} \text{ even integer}$

$$x + 2(x + 2) = x + 4 + 20$$

$$3x + 4 = x + 24$$

$$2x + 4 = 20$$

$$2x = 16$$

$$x = 8$$

$$\boxed{8, 10, 12}$$

Name: _____

Unit 1: Equations & Inequalities

Date: _____ Per: _____

Homework 3: Solving Equations

** This is a 2-page document! **

Solve each equation.

1. $-5(3x - 4) + 4 = 114$

$$-15x + 20 + 4 = 114$$

$$-15x + 24 = 114$$

$$-15x = 90$$

$$x = -6$$

2. $-3 = \frac{9(5-2k)}{3}$

$$-9 = 45 - 18k$$

$$-54 = -18k$$

$$3 = k$$

3. $\left[\frac{15}{8}m - 11 = \frac{7}{4}m + 1\right] \cdot 8$

$$15m - 88 = 14m + 8$$

$$m - 88 = 8$$

$$m = 96$$

4. $-8\left(3x - \frac{1}{4}\right) = 2(9 - 7x)$

$$-24x + 2 = 18 - 14x$$

$$-10x + 2 = 18$$

$$-10x = 16$$

$$x = -\frac{8}{5}$$

5. $\frac{4}{9}\left(\frac{3}{8}x - \frac{27}{16}\right) = \frac{13}{4}$

$$\frac{12}{72}x - \frac{108}{144} = \frac{13}{4}$$

$$\frac{1}{6}x - \frac{3}{4} = \frac{13}{4}$$

$$\frac{1}{6}x = 4$$

$$x = 24$$

6. $10 - (8p + 3) = 9(2p - 5)$

$$10 - 8p - 3 = 18p - 45$$

$$7 - 8p = 18p - 45$$

$$7 = 26p - 45$$

$$52 = 26p$$

$$2 = p$$

7. $3(n + 11) - 5n = -2(n - 12) + 9$

$$3n + 33 - 5n = -2n + 24 + 9$$

$$-2n + 33 = -2n + 33$$

$$33 = 33$$

$$\mathbb{R}; \text{ all real numbers}$$

8. $-9 - (18v - 6) = 2(7v - 16) - 3v$

$$-9 - 18v + 6 = 14v - 32 - 3v$$

$$-3 - 18v = 11v - 32$$

$$-3 = 29v - 32$$

$$29 = 29v$$

$$1 = v$$

$$\begin{aligned}
 9. -8 - 3(w + 13) &= 4(w + 11) - 7w \\
 -8 - 3w - 39 &= 4w + 44 - 7w \\
 -3w - 47 &= -3w + 44 \\
 -47 &\neq 44
 \end{aligned}$$

$\emptyset$; No solution

$$\begin{aligned}
 10. \frac{3a-1}{8} &= \frac{a+9}{4} \\
 4(3a-1) &= 8(a+9) \\
 12a-4 &= 8a+72 \\
 4a-4 &= 72 \\
 4a &= 76 \\
 a &= 19
 \end{aligned}$$

$$\begin{aligned}
 11. 4c+9 &= \frac{8c+1-(2c-11)}{3} \\
 12c+27 &= 8c+1-2c+11 \\
 12c+27 &= 6c+12 \\
 6c+27 &= 12 \\
 6c &= -15 \\
 c &= -\frac{5}{2}
 \end{aligned}$$

$$\begin{aligned}
 12. \frac{5x+6}{3x-9} &= \frac{7}{3} \\
 3(5x+6) &= 7(3x-9) \\
 15x+18 &= 21x-63 \\
 18 &= 6x-63 \\
 81 &= 6x \\
 \frac{27}{2} &= x
 \end{aligned}$$

Write and solve an equation to solve the problem.

13. The measure of one angle is 8 degrees more than one-third the measure of another other angle. If the angles are supplementary, find both angle measures.

let x = angle
let $\frac{1}{3}x + 8$ = other angle

$$\begin{aligned}
 x + \frac{1}{3}x + 8 &= 180 \\
 \frac{4}{3}x + 8 &= 180 \\
 \frac{4}{3}x &= 172 & 4x &= 516 \\
 x &= 129
 \end{aligned}$$

129°
 51°

14. Find three consecutive odd integers such that the sum of the smallest number and four times the largest number is fifty-one more than twice the median number.

let x = 1st odd int.
let $x+2$ = 2nd odd int.
let $x+4$ = 3rd odd int.

$$\begin{aligned}
 x + 4(x+4) &= 2(x+2) + 51 \\
 5x + 16 &= 2x + 55 \\
 3x &= 39 \\
 x &= 13
 \end{aligned}$$

$13, 15, 17$

Solve each equation for the indicated variable.

$$\begin{aligned}
 15. A &= \frac{1}{2}\pi r^2 & (\text{for } \pi) \\
 2A &= \pi r^2 \\
 \frac{2A}{r^2} &= \pi
 \end{aligned}$$

$$\begin{aligned}
 16. \frac{x-y}{2z} &= 3 & (\text{for } x) \\
 x-y &= 6z \\
 x &= 6z + y
 \end{aligned}$$

$$\begin{aligned}
 17. 4m-n &= 1 & (\text{for } n) \\
 4m &= 1+n \\
 4m-1 &= n
 \end{aligned}$$

$$\begin{aligned}
 18. 9a &= \frac{12a+3b}{4} & (\text{for } b) \\
 36a &= 12a+3b \\
 24a &= 3b \\
 8a &= b
 \end{aligned}$$

Name:		Date:	
Topic:		Class:	

Main Ideas/Questions	Notes
WARM-UP	1. If $ x = 7$, what are the possible values of x ? 7 or -7
	2. If $ x = -4$, what are the possible values of x ? None; x cannot be -4 away from zero on the number line.
Steps to solve ABSOLUTE VALUE EQUATIONS	① ISOLATE the absolute value expression.
	② CREATE TWO CASES. Set the "inside" equal to both the positive and negative value of the number on the opposite side of the equal sign.
	③ SOLVE both equations.
	④ CHECK for extraneous solutions.
What are Extraneous Solutions?	Solutions that cannot work when plugged into the original equation.
EXAMPLES	Solve each equation. Be sure to check for extraneous solutions.
	3. $x-9 =5$ $x-9=5$ $x-9=-5$ $x=14$ ✓ $x=4$ ✓ $x = \{4, 14\}$ 4. $3= 2k+7$ $2k+7=3$ $2k+7=-3$ $2k=-4$ $2k=-10$ $k=-2$ ✓ $k=-5$ ✓ $k = \{-5, -2\}$
	5. $y-1 +17=23$ $y-1 =6$ $y-1=6$ $y-1=-6$ $y=7$ ✓ $y=-5$ ✓ $y = \{-5, 7\}$ 6. $3 m+8 =39$ $m+8 =13$ $m+8=13$ $m+8=-13$ $m=5$ $m=-21$ $m = \{-21, 5\}$
	7. $16 r-5 =-32$ $r-5 =-2$ No Solution 8. $\frac{ y-10 }{-2}=-6$ $y-10 =12$ $y-10=12$ $y-10=-12$ $y=22$ ✓ $y=-2$ ✓ $y = \{-2, 22\}$

$$9. -5 = |8 - p| - 11$$

$$6 = |8 - p|$$

$$6 = 8 - p$$

$$-6 = 8 - p$$

$$-2 = -p$$

$$-14 = -p$$

$$p = 2 \checkmark$$

$$p = 14 \checkmark$$

$$p = \{2, 14\}$$

$$10. \frac{|3a + 12|}{9} = 0$$

$$|3a + 12| = 0$$

$$3a + 12 = 0$$

$$3a = -12$$

$$a = -4 \checkmark$$

$$a = \{-4\}$$

$$11. 9 - 4|c + 10| = -15$$

$$-4|c + 10| = -24$$

$$|c + 10| = 6$$

$$c + 10 = 6$$

$$c + 10 = -6$$

$$c = -4 \checkmark$$

$$c = -16 \checkmark$$

$$c = \{-16, -4\}$$

$$12. -16 = -12\left|\frac{k}{5}\right| + 8$$

$$-24 = -12\left|\frac{k}{5}\right|$$

$$2 = \left|\frac{k}{5}\right|$$

$$\frac{k}{5} = 2$$

$$\frac{k}{5} = -2$$

$$k = 10 \checkmark$$

$$k = -10 \checkmark$$

$$k = \{-10, 10\}$$

$$13. |3x - 9| = x + 1$$

$$3x - 9 = x + 1$$

$$3x - 9 = -x - 1$$

$$2x - 9 = 1$$

$$4x - 9 = -1$$

$$2x = 10$$

$$4x = 8$$

$$x = 5 \checkmark$$

$$x = 2 \checkmark$$

$$x = \{2, 5\}$$

$$14. 2|h + 8| = 14h$$

$$|h + 8| = 7h$$

$$h + 8 = 7h$$

$$h + 8 = -7h$$

$$8 = 6h$$

$$8 = -8h$$

$$\frac{4}{3} = h \checkmark$$

$$-1 = h$$

$$h = \{\frac{4}{3}\}$$

$$15. |8 - 2n| - 5n = n + 2$$

$$|8 - 2n| = 6n + 2$$

$$8 - 2n = 6n + 2$$

$$8 - 2n = -6n - 2$$

$$8 = 8n + 2$$

$$8 = -4n - 2$$

$$6 = 8n$$

$$10 = -4n$$

$$\frac{3}{4} = n$$

$$\cancel{-\frac{5}{2} = n}$$

$$n = \{\frac{3}{4}\}$$

$$16. -4w - 2 = |6w + 20|$$

$$-4w - 2 = 6w + 20$$

$$4w + 2 = 6w + 20$$

$$-2 = 10w + 20$$

$$2 = 2w + 20$$

$$-22 = 10w$$

$$-18 = 2w$$

$$-\frac{11}{5} = w \checkmark$$

$$-9 = w \checkmark$$

$$w = \{-9, -\frac{11}{5}\}$$

Name: _____

Unit 1: Equations & Inequalities

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Homework 4: Absolute Value Equations

** This is a 2-page document! **

Solve each equation. Be sure to check for extraneous solutions.

1. $|x-7|=16$

$x-7=16$

$x=23 \checkmark$

$x-7=-16$

$x=-9 \checkmark$

$x = \{-9, 23\}$

2. $|-8k|=24$

$-8k=24$

$k=-3 \checkmark$

$-8k=-24$

$k=3 \checkmark$

$k = \{-3, 3\}$

3. $2=|8p+10|$

$8p+10=2$

$8p=-8$

$p=-1 \checkmark$

$8p+10=-2$

$8p=-12$

$p=-\frac{3}{2} \checkmark$

$p = \{-\frac{3}{2}, -1\}$

4. $|\frac{y+9}{4}|=5$

$\frac{y+9}{4}=5$

$y+9=20$

$y=11 \checkmark$

$\frac{y+9}{4}=-5$

$y+9=-20$

$y=-29 \checkmark$

$y = \{-29, 11\}$

5. $-9|c+8|=-99$

$|c+8|=11$

$c+8=11$

$c=3 \checkmark$

$c+8=-11$

$c=-19 \checkmark$

$c = \{-19, 3\}$

6. $|2s|+19=13$

$|2s|=-6$

No Solution!

7. $|2+7v|+11=27$

$|2+7v|=16$

$2+7v=16$

$7v=14$

$v=2 \checkmark$

$2+7v=-16$

$7v=-18$

$v=-\frac{18}{7} \checkmark$

$v = \{-\frac{18}{7}, 2\}$

8. $-6|8x-4|=-72$

$|8x-4|=12$

$8x-4=12$

$8x=16$

$x=2 \checkmark$

$8x-4=-12$

$8x=-8$

$x=-1 \checkmark$

$x = \{-1, 2\}$

$$9. 53 = 5|p-4| + 8$$

$$45 = 5|p-4|$$

$$9 = |p-4|$$

$$9 = p-4$$

$$13 = p \checkmark$$

$$-9 = p-4$$

$$-5 = p \checkmark$$

$$p = \{-5, 13\}$$

$$10. 8 - 12|k+5| = -4$$

$$-12|k+5| = -12$$

$$|k+5| = 1$$

$$k+5 = 1$$

$$k = -4 \checkmark$$

$$k+5 = -1$$

$$k = -6 \checkmark$$

$$k = \{-6, -4\}$$

$$11. -51 = 3 - |6-12r|$$

$$-54 = -|6-12r|$$

$$54 = |6-12r|$$

$$54 = 6-12r$$

$$48 = -12r$$

$$-4 = r \checkmark$$

$$-54 = 6-12r$$

$$-60 = -12r$$

$$5 = r \checkmark$$

$$r = \{-4, 5\}$$

$$12. \frac{|6-2m|}{-4} - 7 = -19$$

$$\frac{|6-2m|}{-4} = -12$$

$$|6-2m| = 48$$

$$6-2m = 48$$

$$-2m = 42$$

$$m = -21 \checkmark$$

$$6-2m = -48$$

$$-2m = -54$$

$$m = 27 \checkmark$$

$$m = \{-21, 27\}$$

$$13. |4x-1| = 2x+13$$

$$4x-1 = 2x+13$$

$$2x-1 = 13$$

$$2x = 14$$

$$x = 7 \checkmark$$

$$4x-1 = -2x-13$$

$$6x-1 = -13$$

$$6x = -12$$

$$x = -2 \checkmark$$

$$x = \{-2, 7\}$$

$$14. 5n-3 = |n+9|$$

$$5n-3 = n+9$$

$$4n-3 = 9$$

$$4n = 12$$

$$n = 3 \checkmark$$

$$-5n+3 = n+9$$

$$-6n+3 = 9$$

$$-6n = 6$$

$$n = -1$$

$$n = \{3\}$$

$$15. z-10 = |7z|-16$$

$$z+6 = |7z|$$

$$z+6 = 7z$$

$$6 = 6z$$

$$1 = z \checkmark$$

$$-z-6 = 7z$$

$$-6 = 8z$$

$$-\frac{3}{4} = z \checkmark$$

$$z = \{-\frac{3}{4}, 1\}$$

$$16. \frac{|12-x|}{2} = 4x$$

$$|12-x| = 8x$$

$$12-x = 8x$$

$$12 = 9x$$

$$\frac{4}{3} = x \checkmark$$

$$12-x = -8x$$

$$12 = -7x$$

$$-\frac{12}{7} = x$$

$$x = \{\frac{4}{3}\}$$

Name: _____

Algebra 2

Date: _____ Per: _____

Unit 1: Equations & Inequalities

Quiz 1-1: Expressions & Equations

Simplify each radical expression.

$$1. \sqrt{54} = \sqrt{9 \cdot 6} \\ = 3\sqrt{6}$$

$$2. -\sqrt{176} = -\sqrt{16 \cdot 11} \\ = -4\sqrt{11}$$

$$3. \sqrt{300} = \sqrt{100 \cdot 3} \\ = 10\sqrt{3}$$

$$4. \sqrt{252} = \sqrt{36 \cdot 7} \\ = 6\sqrt{7}$$

1. $\underline{3\sqrt{6}}$

2. $\underline{-4\sqrt{11}}$

3. $\underline{10\sqrt{3}}$

4. $\underline{6\sqrt{7}}$

Name all sets to which each value belongs. (Use R, I, Q, Z, W, N)

5. $\sqrt{32} = \sqrt{16 \cdot 2} = 4\sqrt{2}$

6. $-\frac{60}{12} = -5$

7. $\pi - \pi = 0$

8. $\sqrt{\frac{81}{4}} = \frac{9}{2}$

5. $\underline{R, I}$

6. $\underline{R, Q, Z}$

7. $\underline{R, Q, Z, W}$

8. $\underline{R, Q}$

Simplify each expression.

$$9. 31 - (5 - 3 \cdot 4)^2 + 2^6 \div 4 \\ = 31 - (5 - 12)^2 + 64 \div 4 \\ = 31 - (-7)^2 + 16 \\ = 31 - 49 + 16 \\ = -18 + 16 \\ = -2$$

$$10. \frac{5 \cdot |-2 - 7(3)| - 2 \cdot 17}{22 + \sqrt{7 \cdot 28}} \\ = \frac{5 \cdot |-2 - 21| - 34}{22 + \sqrt{196}} \\ = \frac{5 \cdot 23 - 34}{22 + 14} \\ = \frac{115 - 34}{36} = \frac{81}{36} = \frac{9}{4}$$

9. $\underline{-2}$

10. $\underline{\frac{9}{4}}$

11. $\underline{-88}$

12. $\underline{\frac{1}{12}}$

$$11. -a^2 + 4ab - b^2 \quad (\text{if } a = 2 \text{ and } b = -6) \\ = -(2)^2 + 4(2)(-6) - (-6)^2 \\ = -4 + 8(-6) - 36 \\ = -4 - 48 - 36 \\ = -52 - 36 \\ = -88$$

$$12. 2x - 5y \quad (\text{if } x = \frac{2}{3} \text{ and } y = \frac{1}{4}) \\ = 2\left(\frac{2}{3}\right) - 5\left(\frac{1}{4}\right) \\ = \frac{4}{3} - \frac{5}{4} \\ = \frac{16}{12} - \frac{15}{12} = \frac{1}{12}$$

Solve each equation.

13. $4x + 29 = 5 - 8(x + 6)$

$$4x + 29 = 5 - 8x - 48$$

$$4x + 29 = -8x - 43$$

$$12x + 29 = -43$$

$$12x = -72$$

$$x = -6$$

14. $-\frac{4}{3}(9 - 2k) = \frac{1}{2}\left(\frac{1}{3}k + 1\right)$

$$\left[-\frac{36}{3} + \frac{8k}{3} = \frac{1}{6}k + \frac{1}{2}\right] \cdot 6$$

$$-72 + 16k = k + 3$$

$$-72 + 15k = 3$$

$$15k = 75$$

$$k = 5$$

13. $x = -6$

14. $k = 5$

15. $m = -\frac{1}{4}$

16. $b = 7a$

15. $\frac{3m-3}{5} = \frac{2m-1}{2}$

$$2(3m-3) = 5(2m-1)$$

$$6m-6 = 10m-5$$

$$-6 = 4m-5$$

$$-1 = 4m$$

$$-\frac{1}{4} = m$$

16. $\frac{11a+b}{3} = 6a$ (solve for b)

$$11a + b = 18a$$

$$b = 7a$$

Solve each equation. Be sure to check for extraneous solutions.

17. $|3 - 6w| = 39$

$$3 - 6w = 39$$

$$-6w = 36$$

$$w = -6 \checkmark$$

$$3 - 6w = -39$$

$$-6w = -42$$

$$w = 7 \checkmark$$

18. $-7 = \frac{|a-3|}{-2}$

$$14 = |a-3|$$

$$14 = a-3$$

$$17 = a \checkmark$$

$$-14 = a-3$$

$$-11 = a \checkmark$$

17. $w = \{-6, 7\}$

18. $a = \{-11, 17\}$

19. $x = \{-1, \frac{1}{2}\}$

20. $p = \{\frac{3}{8}\}$

19. $7|12x+3| - 1 = 62$

$$7|12x+3| = 63$$

$$|12x+3| = 9$$

$$12x+3 = 9$$

$$12x = 6$$

$$x = \frac{1}{2} \checkmark$$

$$12x+3 = -9$$

$$12x = -12$$

$$x = -1 \checkmark$$

20. $|5-3p| + 9 = 13p + 8$

$$|5-3p| = 13p - 1$$

$$5-3p = 13p-1$$

$$5 = 16p-1$$

$$6 = 16p$$

$$\frac{3}{8} = p \checkmark$$

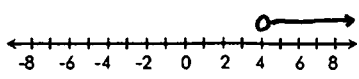
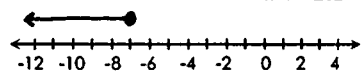
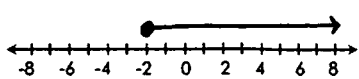
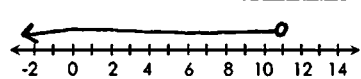
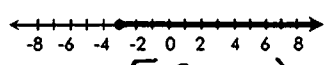
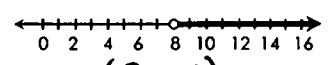
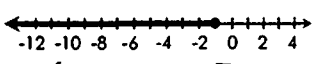
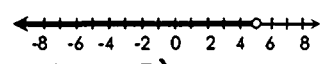
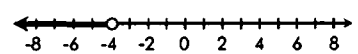
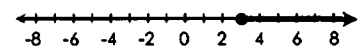
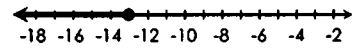
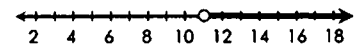
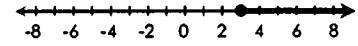
$$5-3p = -13p+1$$

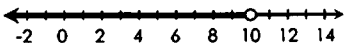
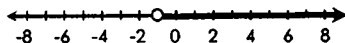
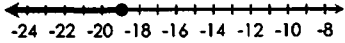
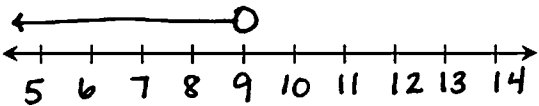
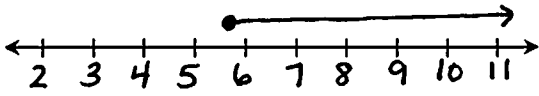
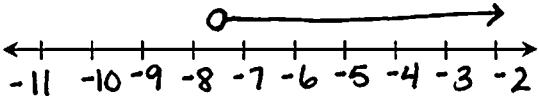
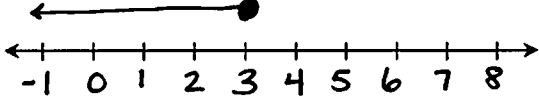
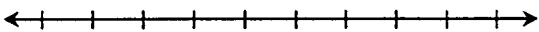
$$5 = -10p+1$$

$$4 = -10p$$

$$-\frac{2}{5} = p$$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples		
Graphing INEQUALITIES	- Use a <u>Closed</u> dot for $\leq$ or $\geq$ symbols. - Use an <u>open</u> dot for $<$ or $>$ symbols.		
	Graph each inequality.		
	1. $x > 4$ 	2. $x \leq -7$ 	
	3. $x \geq -2$ 	4. $x < 11$ 	
INTERVAL & SET Notation	Interval Notation is a way to write the solution to an inequality using <u>infinity symbols</u> , <u>parentheses</u> , and <u>brackets</u> .		
	- Parentheses indicate "not included" or "open" - Brackets indicate "included" or "closed"		
	Write each inequality in interval notation.		
	5.  $[-3, \infty)$	6.  $(8, \infty)$	
	7.  $(-\infty, -1]$	8.  $(-\infty, 5)$	
Set Notation is another way to write the solution to an inequality.			
Example: $x > 5$ Set Notation: $\{x \mid x > 5\}$			
Read as <u>the set of x-values such that x is greater than 5</u>			
WRITING Inequalities	Write the inequality shown on the graph in interval and set notation.		
	Graph	Interval Notation	Set Notation
	1. 	$(-\infty, -4)$	$\{x \mid x < -4\}$
	2. 	$[3, \infty)$	$\{x \mid x \geq 3\}$
	3. 	$(-\infty, -13]$	$\{x \mid x \leq -13\}$
	4. 	$(11, \infty)$	$\{x \mid x > 11\}$
	5. 	$[3, \infty)$	$\{x \mid x \geq 3\}$

SOLVING Multi-Step Inequalities WATCH OUT: When you multiply or divide by a negative, Switch the inequality direction.	6.		$(-\infty, 10)$	$\{x \mid x < 10\}$
	7.		$(-1, \infty)$	$\{x \mid x > -1\}$
	8.		$(-\infty, -19]$	$\{x \mid x \leq -19\}$
	Solve, graph, and write each solution in INTERVAL NOTATION.			
	9.	$5v + 1 > 7v - 17$ $1 > 2v - 17$ $18 > 2v$ $9 > v$ $v < 9$	 Interval Notation: $(-\infty, 9)$	
	10.	$21x - 3(5x + 9) \geq 7$ $21x - 15x - 27 \geq 7$ $6x - 27 \geq 7$ $6x \geq 34$ $x \geq \frac{17}{3}$	 Interval Notation: $[\frac{17}{3}, \infty)$	
	11.	$12p + 52 > -2(4 - 2p)$ $12p + 52 > -8 + 4p$ $8p + 52 > -8$ $8p > -60$ $p > -7.5$	 Interval Notation: $(-7.5, \infty)$	
	12.	$3 - (4m + 1) \geq 5m - 25$ $3 - 4m - 1 \geq 5m - 25$ $2 - 4m \geq 5m - 25$ $2 \geq 9m - 25$ $27 \geq 9m$ $3 \geq m ; m \leq 3$	 Interval Notation: $(-\infty, 3]$	
	13.	$3(8x - 9) > 4(6x + 1)$ $24x - 27 > 24x + 4$ $-27 > 4$ No Solution	 Interval Notation: $\emptyset$	

Solve, graph, and write each solution in SET NOTATION.

14. $10(a-3) - 13a \leq 9(a-8)$

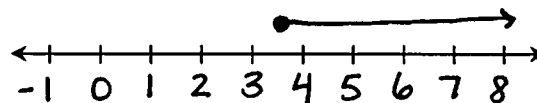
$$10a - 30 - 13a \leq 9a - 72$$

$$-3a - 30 \leq 9a - 72$$

$$-12a - 30 \leq -72$$

$$-12a \leq -42$$

$$a \geq 3.5$$



Set Notation: $\{a | a \geq 3.5\}$

15. $8 - 7(2r + 3) \leq 3(r - 10)$

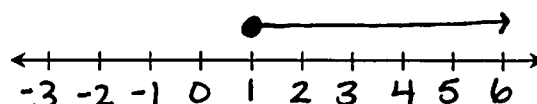
$$8 - 14r - 21 \leq 3r - 30$$

$$-14r - 13 \leq 3r - 30$$

$$-17r - 13 \leq -30$$

$$-17r \leq -17$$

$$r \geq 1$$



Set Notation: $\{r | r \geq 1\}$

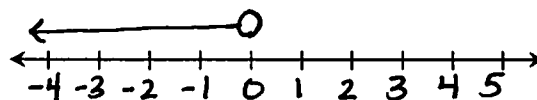
16. $-3(2p + 8) < -4(2p + 6)$

$$-6p - 24 < -8p - 24$$

$$2p - 24 < -24$$

$$2p < 0$$

$$p < 0$$



Set Notation: $\{p | p < 0\}$

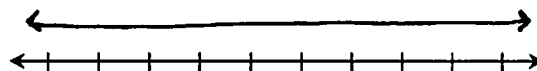
17. $11x - 4 \leq 6x - (3 - 5x)$

$$11x - 4 \leq 6x - 3 + 5x$$

$$11x - 4 \leq 11x - 3$$

$$-4 \leq -3$$

$$\mathbb{R}$$



Set Notation: $\{x | x \in \mathbb{R}\}$

18. $\frac{4 + 5(2w - 10)}{-2} < w + 7$

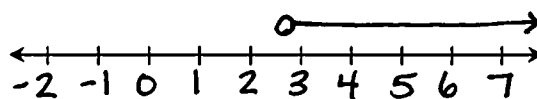
$$4 + 10w - 50 > -2w - 14$$

$$10w - 46 > -2w - 14$$

$$12w - 46 > -14$$

$$12w > 32$$

$$w > \frac{8}{3}$$



Set Notation: $\{w | w > \frac{8}{3}\}$

Name: _____

Unit 1: Equations & Inequalities

Date: _____ Per: _____

Homework 5: Multi-Step Inequalities

**** This is a 2-page document! ******Solve, graph, and write the solution to each inequality in both INTERVAL and SET NOTATION.**

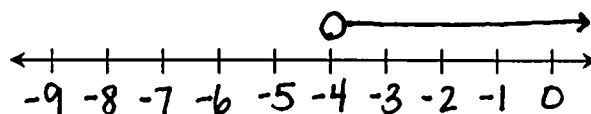
1. $-7a - 11 < 1 - 4a$

$-3a - 11 < 1$

$-3a < 12$

$a > -4$

Graph:



Interval Notation:

$(-4, \infty)$

Set Notation:

$\{a \mid a > -4\}$

2. $\frac{3}{4}(12v - 28) - v < -5$

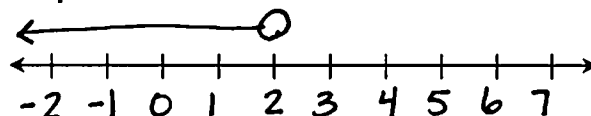
$9v - 21 - v < -5$

$8v - 21 < -5$

$8v < 16$

$v < 2$

Graph:



Interval Notation:

$(-\infty, 2)$

Set Notation:

$\{v \mid v < 2\}$

3. $5 - 9(8 - 2x) \leq 2x - 115$

$5 - 72 + 18x \leq 2x - 115$

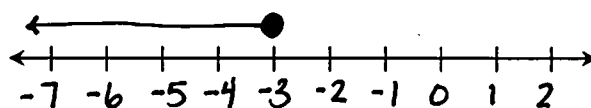
$-67 + 18x \leq 2x - 115$

$-67 + 16x \leq -115$

$16x \leq -48$

$x \leq -3$

Graph:



Interval Notation:

$(-\infty, -3]$

Set Notation:

$\{x \mid x \leq -3\}$

4. $-2(7 + y) \geq -8(y + 1)$

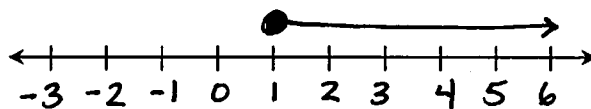
$-14 - 2y \geq -8y - 8$

$-14 + 6y \geq -8$

$6y \geq 6$

$y \geq 1$

Graph:



Interval Notation:

$[1, \infty)$

Set Notation:

$\{y \mid y \geq 1\}$

5. $-5(k + 4) > 2 - (3k + 6)$

$-5k - 20 > 2 - 3k - 6$

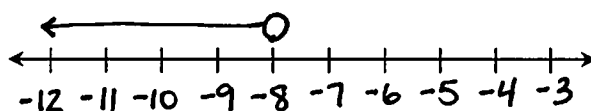
$-5k - 20 > -3k - 4$

$-2k - 20 > -4$

$-2k > 16$

$k < -8$

Graph:

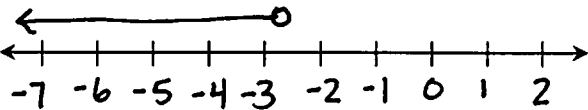
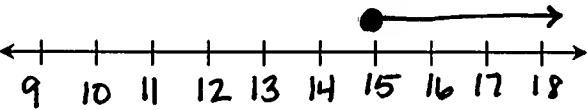
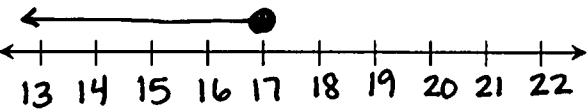
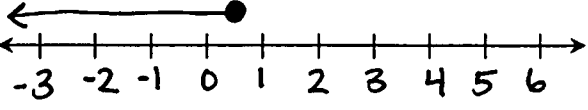


Interval Notation:

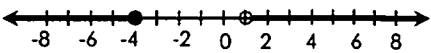
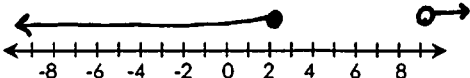
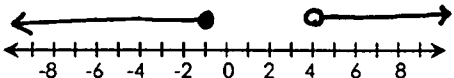
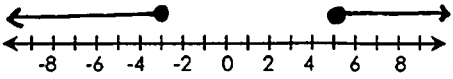
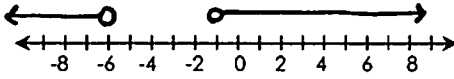
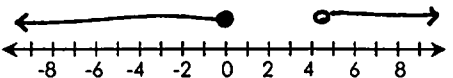
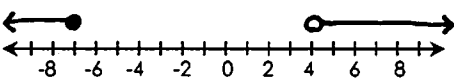
$(-\infty, -8)$

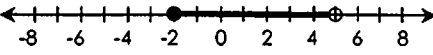
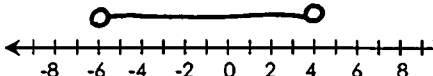
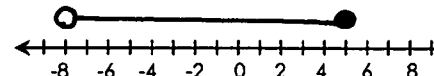
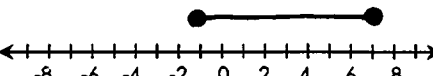
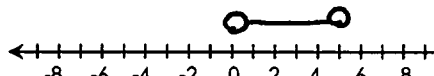
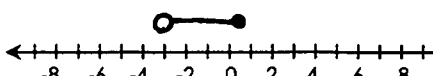
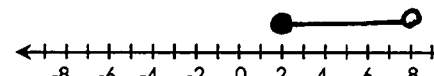
Set Notation:

$\{k \mid k < -8\}$

6. $\frac{3(2x+7)}{-3} > x+1$ $6x+21 < -3x-3$ $9x+21 < -3$ $9x < -24$ $x < -\frac{8}{3}$	Graph:  <table border="1" data-bbox="888 411 1480 527"> <tr> <td>Interval Notation:</td><td>Set Notation:</td></tr> <tr> <td>$(-\infty, -\frac{8}{3})$</td><td>$\{x \mid x < -\frac{8}{3}\}$</td></tr> </table>	Interval Notation:	Set Notation:	$(-\infty, -\frac{8}{3})$	$\{x \mid x < -\frac{8}{3}\}$
Interval Notation:	Set Notation:				
$(-\infty, -\frac{8}{3})$	$\{x \mid x < -\frac{8}{3}\}$				
7. $\frac{2}{3}p - 2(4-p) \geq 32$ $\frac{2}{3}p - 8 + 2p \geq 32$ $2p - 24 + 6p \geq 96$ $8p \geq 120$ $p \geq 15$	Graph:  <table border="1" data-bbox="888 774 1480 890"> <tr> <td>Interval Notation:</td><td>Set Notation:</td></tr> <tr> <td>$[15, \infty)$</td><td>$\{p \mid p \geq 15\}$</td></tr> </table>	Interval Notation:	Set Notation:	$[15, \infty)$	$\{p \mid p \geq 15\}$
Interval Notation:	Set Notation:				
$[15, \infty)$	$\{p \mid p \geq 15\}$				
8. $r+10 \leq -3(2r-3) + 6(r+3)$ $r+10 \leq -6r+9+6r+18$ $r+10 \leq 27$ $r \leq 17$	Graph:  <table border="1" data-bbox="888 1138 1480 1253"> <tr> <td>Interval Notation:</td><td>Set Notation:</td></tr> <tr> <td>$(-\infty, 17]$</td><td>$\{r \mid r \leq 17\}$</td></tr> </table>	Interval Notation:	Set Notation:	$(-\infty, 17]$	$\{r \mid r \leq 17\}$
Interval Notation:	Set Notation:				
$(-\infty, 17]$	$\{r \mid r \leq 17\}$				
9. $-8m-12 > -5(m-8) - 3(m-7)$ $-8m-12 > -5m+40-3m+21$ $-8m-12 > -8m+61$ $-12 > 61$ No Solution!	Graph:  <table border="1" data-bbox="888 1499 1480 1614"> <tr> <td>Interval Notation:</td><td>Set Notation:</td></tr> <tr> <td>$\emptyset$</td><td>$\{ \}$</td></tr> </table>	Interval Notation:	Set Notation:	$\emptyset$	$\{ \}$
Interval Notation:	Set Notation:				
$\emptyset$	$\{ \}$				
10. $18\left(q-\frac{3}{4}\right) \leq -2\left(q+\frac{7}{4}\right)$ $18q - \frac{27}{2} \leq -2q - \frac{7}{2}$ $20q - \frac{27}{2} \leq -\frac{7}{2}$ $20q \leq 10$ $q \leq \frac{1}{2}$	Graph:  <table border="1" data-bbox="888 1864 1480 1980"> <tr> <td>Interval Notation:</td><td>Set Notation:</td></tr> <tr> <td>$(-\infty, \frac{1}{2}]$</td><td>$\{q \mid q \leq \frac{1}{2}\}$</td></tr> </table>	Interval Notation:	Set Notation:	$(-\infty, \frac{1}{2}]$	$\{q \mid q \leq \frac{1}{2}\}$
Interval Notation:	Set Notation:				
$(-\infty, \frac{1}{2}]$	$\{q \mid q \leq \frac{1}{2}\}$				

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes	
COMPOUND <i>Inequalities</i>	Two or more inequalities graphed together on the same number line.	
TYPE 1: "OR"		Inequality: $x \leq -4$ or $x > 1$ Interval Notation: $(-\infty, -4] \cup (1, \infty)$ Set Notation: $\{x \mid x \leq -4 \text{ or } x > 1\}$
SOLVING "OR" <i>Inequalities</i> To solve "or" inequalities, solve each part, then graph on a number line to show the solutions.	Solve, graph, and write each solution in interval notation.	
	1. $3x + 4 \leq 10$ or $5x > 45$ $3x \leq 6$ $x > 9$ $x \leq 2$ $x \leq 2$ or $x > 9$  Interval Notation: $(-\infty, 2] \cup (9, \infty)$	2. $n + 6 \leq 5$ or $9n + 7 > 43$ $n \leq -1$ $9n > 36$ $n > 4$ $n \leq -1$ or $n > 4$  Interval Notation: $(-\infty, -1] \cup (4, \infty)$
	3. $-3y + 8 \leq -7$ or $10y - 3 \leq -33$ $-3y \leq -15$ $10y \leq -30$ $y \geq 5$ $y \leq -3$ $y \leq -3$ or $y \geq 5$  Interval Notation: $(-\infty, -3] \cup [5, \infty)$	4. $6 - 2a < 8$ or $\frac{a}{2} + 1 < -2$ $-2a < 2$ $\frac{a}{2} < -3$ $a > -1$ $a < -6$ $a < -6$ or $a > -1$  Interval Notation: $(-\infty, -6) \cup (-1, \infty)$
	5. $3(5 - 2k) < -12$ or $\frac{1}{3}(15k - 9) \leq -3$ $15 - 6k < -12$ $5k - 3 \leq -3$ $-6k < -27$ $5k \leq 0$ $k > 4.5$ $k \leq 0$ $k \leq 0$ or $k > 4.5$  Interval Notation: $(-\infty, 0] \cup (4.5, \infty)$	6. $\frac{8-x}{-3} \leq -5$ or $5 - \frac{3}{2}x < -1$ $8 - x \geq 15$ $-\frac{3}{2}x < -6$ $-x \geq 7$ $x > 4$ $x \leq -7$ $x > 4$ $x \leq -7$ or $x > 4$  Interval Notation: $(-\infty, -7] \cup (4, \infty)$

TYPE 2: "AND"		Inequality: $x \geq -2$ and $x < 5$ Interval Notation: $[-2, 5)$ Set Notation: $\{x \mid x \geq -2 \text{ and } x < 5\}$
REWRITING "AND" Inequalities	Because the solutions to an "and" inequality fall between two endpoints, they are frequently written in a more condensed form. Example: $x \geq -2$ and $x < 5 \Rightarrow -2 \leq x < 5$	
SOLVING "AND" <i>Inequalities</i> If the "and" inequality is written out, solve each part separately. If condensed, you can solve it all together, working inside out. Then, graph to show all possible solutions.	Solve, graph, and write each solution in interval notation. 7. $2x + 3 > -9$ and $8x - 2 < 30$ $2x > -12$ $8x < 32$ $x > -6$ $x < 4$ $-6 < x < 4$  Interval Notation: $(-6, 4)$ 8. $2w + 1 \leq 11$ and $1 - 5w < 41$ $2w \leq 10$ $-5w < 40$ $w \leq 5$ $w > -8$ $-8 < w \leq 5$  Interval Notation: $(-8, 5]$ 9. $-2 \leq a - 1 \leq 6$ $-1 \leq a \leq 7$  Interval Notation: $[-1, 7]$ 10. $4 < 3p + 4 < 19$ $0 < 3p < 15$ $0 < p < 5$  Interval Notation: $(0, 5)$ 11. $28 > -2 - 10m \geq -7$ $30 > -10m \geq -5$ $-3 < m \leq \frac{1}{2}$  Interval Notation: $(-3, \frac{1}{2}]$ 12. $-5 \leq \frac{12-x}{-2} < -2$ $10 \geq 12 - x > 4$ $-2 \geq -x > -8$ $2 \leq x < 8$  Interval Notation: $[2, 8)$	

Name:	Date:
Topic:	Class:

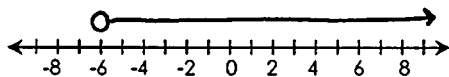
Main Ideas/Questions	Notes																											
Special Cases of COMPOUND INEQUALITIES	We previously learned that compound inequalities are two inequalities joined by the word "or" or "and". Now we will be looking at special cases of compound inequalities in which there is no solution, infinitely many solutions, or overlapping solutions.																											
	Example at the Zoo: A sign at the ticket booth reads "Guests that are under 3 years old and over 62 years old get free admission!" Does this make sense? If not, why? How would you change it? Explain.																											
	<u>* Answers may vary *</u>																											
	<u>No - you cannot be under 3 years old and over 62 years old at the same time.</u>																											
	<u>The sign should say "or" instead of "and."</u>																											
	Graph the inequalities below and give your answer in interval notation.																											
	<table><thead><tr><th>Inequality</th><th>Graph</th><th>Interval Notation</th></tr></thead><tbody><tr><td>1. $x < -3$ and $x > 7$</td><td></td><td>$\emptyset$</td></tr><tr><td>2. $a \geq 0$ and $a < 6$</td><td></td><td>$[0, 6)$</td></tr><tr><td>3. $k \leq 5$ and $k \leq 8$</td><td></td><td>$(-\infty, 5]$</td></tr><tr><td>4. $m > -4$ and $m \geq 2$</td><td></td><td>$[2, \infty)$</td></tr><tr><td>5. $r < -8$ or $r > 1$</td><td></td><td>$(-\infty, -8) \cup (1, \infty)$</td></tr><tr><td>6. $y \geq -2$ or $y > 7$</td><td></td><td>$[-2, \infty)$</td></tr><tr><td>7. $n \leq 3$ or $n \leq -9$</td><td></td><td>$(-\infty, 3]$</td></tr><tr><td>8. $b > -1$ or $b < 1$</td><td></td><td>$(-\infty, \infty)$</td></tr></tbody></table>	Inequality	Graph	Interval Notation	1. $x < -3$ and $x > 7$		$\emptyset$	2. $a \geq 0$ and $a < 6$		$[0, 6)$	3. $k \leq 5$ and $k \leq 8$		$(-\infty, 5]$	4. $m > -4$ and $m \geq 2$		$[2, \infty)$	5. $r < -8$ or $r > 1$		$(-\infty, -8) \cup (1, \infty)$	6. $y \geq -2$ or $y > 7$		$[-2, \infty)$	7. $n \leq 3$ or $n \leq -9$		$(-\infty, 3]$	8. $b > -1$ or $b < 1$		$(-\infty, \infty)$
	Inequality	Graph	Interval Notation																									
	1. $x < -3$ and $x > 7$		$\emptyset$																									
	2. $a \geq 0$ and $a < 6$		$[0, 6)$																									
3. $k \leq 5$ and $k \leq 8$		$(-\infty, 5]$																										
4. $m > -4$ and $m \geq 2$		$[2, \infty)$																										
5. $r < -8$ or $r > 1$		$(-\infty, -8) \cup (1, \infty)$																										
6. $y \geq -2$ or $y > 7$		$[-2, \infty)$																										
7. $n \leq 3$ or $n \leq -9$		$(-\infty, 3]$																										
8. $b > -1$ or $b < 1$		$(-\infty, \infty)$																										

Solve, then graph the solutions and give your answer in interval notation.

9. $x - 4 > -1$ or $-8x < 48$

$x > 3$ $x > -6$

$x > -6$ or $x > 3$

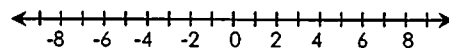


Interval Notation: $(-6, \infty)$

10. $\frac{p}{8} \leq -1$ and $p + 9 > 15$

$p \leq -8$ $p > 6$

$p \leq -8$ and $p > 6$; $\emptyset$



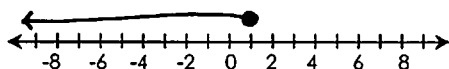
Interval Notation: $\emptyset$

11. $7 + r < 10$ and $-2r - 1 \geq -3$

$r < 3$ $-2r \geq -2$

$r \leq 1$

$r < 3$ and $r \leq 1$; $r \leq 1$



Interval Notation: $(-\infty, 1]$

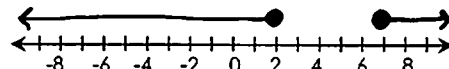
12. $\frac{a-4}{-2} \geq 1$ or $a - 1 \geq 6$

$a - 4 \leq -2$

$a \geq 7$

$a \leq 2$

$a \leq 2$ or $a \geq 7$



Interval Notation: $(-\infty, 2] \cup [7, \infty)$

13. $5k + 3 > 8$ and $9k + 8 \geq -73$

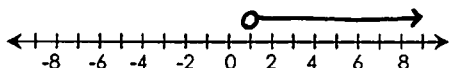
$5k > 5$

$9k \geq -81$

$k > 1$

$k \geq -9$

$k \geq -9$ and $k > 1$; $k > 1$



Interval Notation: $(1, \infty)$

14. $1 - 2y \geq -17$ or $5y - 5 \leq -50$

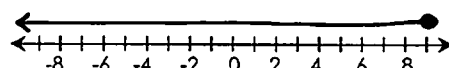
$-2y \geq -18$

$5y \leq -45$

$y \leq 9$

$y \leq -9$

$y \leq -9$ or $y \leq 9$; $y \leq 9$



Interval Notation: $(-\infty, 9]$

15. $2x + 6 > -4$ or $9x + 5 \leq 41$

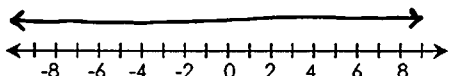
$2x > -10$

$9x \leq 36$

$x > -5$

$x \leq 4$

$x > -5$ or $x \leq 4$; $\mathbb{R}$



Interval Notation: $(-\infty, \infty)$

16. $10v + 4 < 44$ and $3v + 6 > -15$

$10v < 40$

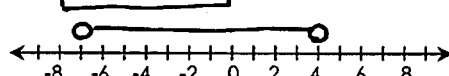
$3v > -21$

$v < 4$

$v > -7$

$v > -7$ and $v < 4$

$-7 < v < 4$



Interval Notation: $(-7, 4)$

17. $10n + 1 \geq -59$ or $-10 - 9n < 26$

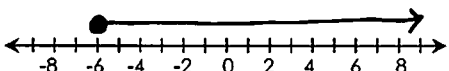
$10n \geq -60$

$-9n < 36$

$n \geq -6$

$n > -4$

$n \geq -6$ or $n > -4$; $n \geq -6$



Interval Notation: $[-6, \infty)$

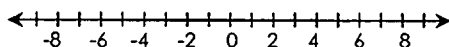
18. $6k - 4 \geq -46$ and $k + 5 \leq -5$

$6k \geq -42$

$k \leq -10$

$k \geq -7$

$k \leq -10$ and $k \geq -7$; $\emptyset$



Interval Notation: $\emptyset$

Name: _____

Unit 1: Equations & Inequalities

Date: _____ Per: _____

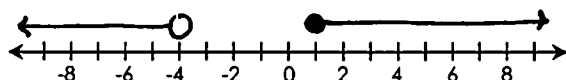
Homework 6: Compound Inequalities

**** This is a 2-page document! ******Solve, graph, and write the solution to each compound inequality in interval notation.**

1. $-2x - 7 > 1$ or $x - 2 \geq -1$

$$\begin{aligned} -2x &> 8 & x &\geq 1 \\ x &< -4 \end{aligned}$$

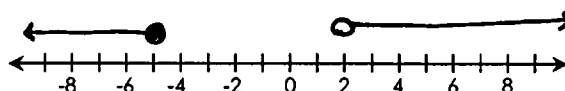
$$x < -4 \text{ or } x \geq 1$$

Interval Notation: $(-\infty, -4) \cup [1, \infty)$

2. $\frac{a}{-2} < -1$ or $-4a + 3 \geq 23$

$$\begin{aligned} a &> 2 & -4a &\geq 20 \\ & & a &\leq -5 \end{aligned}$$

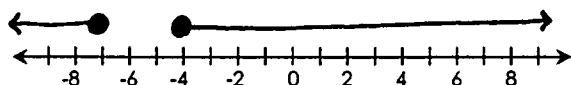
$$a \leq -5 \text{ or } a > 2$$

Interval Notation: $(-\infty, -5] \cup (2, \infty)$

3. $6v + 38 \leq -4$ or $2(v + 3) \geq -2$

$$\begin{aligned} 6v &\leq -42 & 2v + 6 &\geq -2 \\ v &\leq -7 & 2v &\geq -8 \\ & & v &\geq -4 \end{aligned}$$

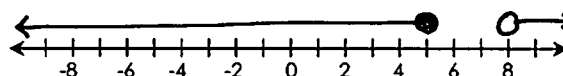
$$v \leq -7 \text{ or } v \geq -4$$

Interval Notation: $(-\infty, -7] \cup [-4, \infty)$

4. $4(1 - k) \geq -16$ or $7 - 6k < -41$

$$\begin{aligned} 4 - 4k &\geq -16 & -6k &< -48 \\ -4k &\geq -20 & k &> 8 \\ k &\leq 5 \end{aligned}$$

$$k \leq 5 \text{ or } k > 8$$

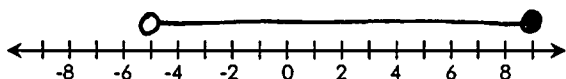
Interval Notation: $(-\infty, 5] \cup (8, \infty)$

5. $10n - 9 > -59$ and $n - 6 \leq 3$

$$\begin{aligned} 10n &> -50 & n &\leq 9 \\ n &> -5 \end{aligned}$$

$$n > -5 \text{ and } n \leq 9$$

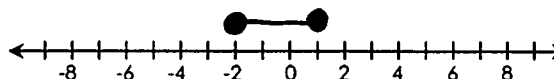
$$-5 < n \leq 9$$

Interval Notation: $(-5, 9]$

6. $-13 \leq 3 + 8p \leq 11$

$$-16 \leq 8p \leq 8$$

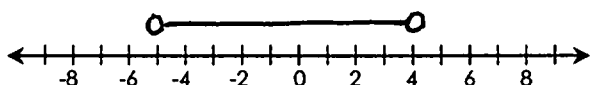
$$-2 \leq p \leq 1$$

Interval Notation: $[-2, 1]$

7. $37 > 7 - 6x > -17$

$$30 > -6x > -24$$

$$-5 < x < 4$$



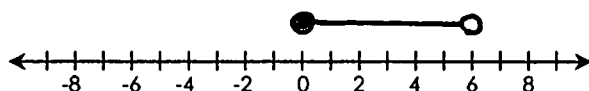
Interval Notation: $(-5, 4)$

8. $3 \geq \frac{9-w}{3} > 1$

$$9 \geq 9-w > 3$$

$$0 \geq -w > -6$$

$$0 \leq w < 6$$



Interval Notation: $[0, 6)$

9. $2r + 6 \leq 4$ and $6r - 3 < 39$

$$2r \leq -2$$

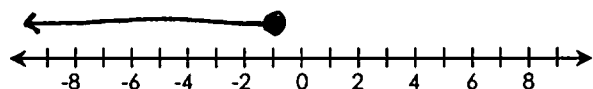
$$6r < 42$$

$$r \leq -1$$

$$r < 7$$

$$r \leq -1 \text{ and } r < 7;$$

$$\boxed{r \leq -1}$$



Interval Notation: $(-\infty, -1]$

10. $6m + 8 > -10$ or $10m + 9 < 39$

$$6m > -18$$

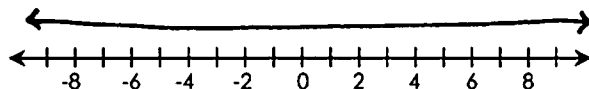
$$10m < 30$$

$$m > -3$$

$$m < 3$$

$$m > -3 \text{ or } m < 3;$$

$$\boxed{\text{TR}}$$



Interval Notation: $(-\infty, \infty)$

11. $6y + 2 < -4$ and $-4(2y + 3) < -68$

$$6y < -6$$

$$-8y - 12 < -68$$

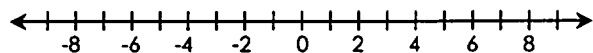
$$y < -1$$

$$-8y < -56$$

$$y > 7$$

$$y < -1 \text{ and } y > 7;$$

$$\boxed{\text{No Solution}}$$



Interval Notation: $\emptyset$

12. $\frac{4}{3}x + 11 \geq 7$ or $-4x - 3 \leq -23$

$$\frac{4}{3}x \geq -4$$

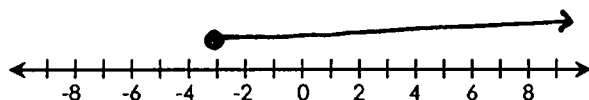
$$-4x \leq -20$$

$$x \geq -3$$

$$x \geq 5$$

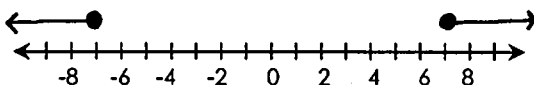
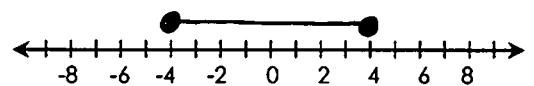
$$x \geq -3 \text{ or } x \geq 5;$$

$$\boxed{x \geq -3}$$

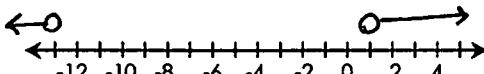
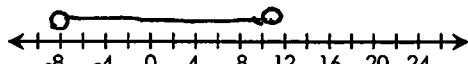
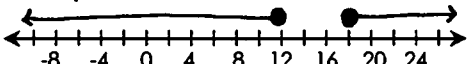
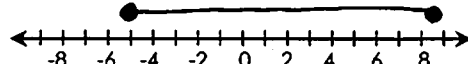


Interval Notation: $[-3, \infty)$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes	
ABSOLUTE VALUE <i>Inequalities</i>	CASE 1	GREATER THAN/GREATER THAN OR EQUAL TO
		Example: $ x \geq 7$ $x \geq 7$ $x \leq -7$  Interval Notation: $(-\infty, -7] \cup [7, \infty)$
	CASE 2	LESS THAN/LESS THAN OR EQUAL TO
		Example: $ x \leq 4$ $x \leq 4$ $x \geq -4$  Interval Notation: $[-4, 4]$
What does this mean?	Absolute value inequalities are compound inequalities, too!	
SOLVING ABSOLUTE VALUE <i>Inequalities</i>	❶	ISOLATE the absolute value expression.
	❷	CREATE TWO CASES. Use the "KISS" Method (Keep it, switch, switch) to set up the two cases.
	❸	SOLVE both inequalities.
	❹	GRAPH your solutions and write your answer in interval notation.

Solve, graph, and write each solution in interval notation.

1. $x+6 > 7$ $x+6 > 7$ $x > 1$ $x+6 < -7$ $x < -13$ $x < -13 \text{ or } x > 1$  Interval Notation: $(-\infty, -13) \cup (1, \infty)$	2. $2v-3 < 19$ $2v-3 < 19$ $2v < 22$ $v < 11$ $2v-3 > -19$ $2v > -16$ $v > -8$ $-8 < v < 11$  Interval Notation: $(-8, 11)$
3. $-5+\frac{k}{3} \geq 1$ $-5+\frac{k}{3} \geq 1$ $\frac{k}{3} \geq 6$ $k \geq 18$ $-5+\frac{k}{3} \leq -1$ $\frac{k}{3} \leq 4$ $k \leq 12$ $k \leq 12 \text{ or } k \geq 18$  Interval Notation: $(-\infty, 12] \cup [18, \infty)$	4. $7-4n \leq 27$ $7-4n \leq 27$ $-4n \leq 20$ $n \geq -5$ $7-4n \geq -27$ $-4n \geq -34$ $n \leq 8.5$ $-5 \leq n \leq 8.5$  Interval Notation: $[-5, 8.5]$

$$5. \frac{|q-3|}{5} \geq 1 \quad |q-3| \geq 5$$

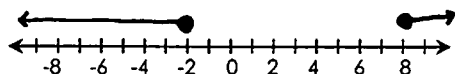
$$q-3 \geq 5$$

$$q \geq 8$$

$$q-3 \leq -5$$

$$q \leq -2$$

$$q \leq -2 \text{ or } q \geq 8$$



Interval Notation: $(-\infty, -2] \cup [8, \infty)$

$$6. |8a+10|+3 < 25 \quad |8a+10| < 22$$

$$8a+10 < 22$$

$$8a < 12$$

$$a < 1.5$$

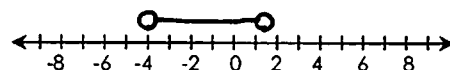
$$8a+10 > -22$$

$$8a > -32$$

$$a > -4$$

$$a > -4 \text{ and } a < 1.5$$

$$-4 < a < 1.5$$



Interval Notation: $(-4, 1.5)$

$$7. -10|2r-1| \leq -60 \quad |2r-1| \geq 6$$

$$2r-1 \geq 6$$

$$2r \geq 7$$

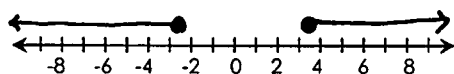
$$r \geq 3.5$$

$$2r-1 \leq -6$$

$$2r \leq -5$$

$$r \leq -2.5$$

$$r \leq -2.5 \text{ or } r \geq 3.5$$



Interval Notation: $(-\infty, -2.5] \cup [3.5, \infty)$

$$8. 9|3-k|-5 \geq 67 \quad 9|3-k| \geq 72$$

$$|3-k| \geq 8$$

$$3-k \geq 8$$

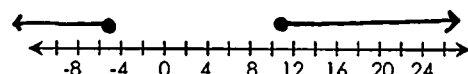
$$-k \geq 5$$

$$k \leq -5$$

$$3-k \leq -8$$

$$-k \leq -11$$

$$k \geq 11$$



Interval Notation: $(-\infty, -5] \cup [11, \infty)$

$$9. 6|-5n|+7 > 37 \quad 6|-5n| > 30$$

$$|-5n| > 5$$

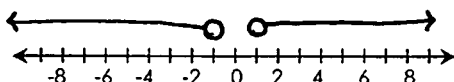
$$-5n > 5$$

$$n < -1$$

$$-5n < -5$$

$$n > 1$$

$$n < -1 \text{ or } n > 1$$



Interval Notation: $(-\infty, -1) \cup (1, \infty)$

$$10. -1-4|x+9| \leq -21 \quad -4|x+9| \leq -20$$

$$|x+9| \geq 5$$

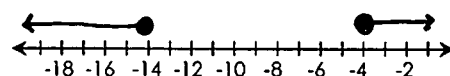
$$x+9 \geq 5$$

$$x \geq -4$$

$$x+9 \leq -5$$

$$x \leq -14$$

$$x \leq -14 \text{ or } x \geq -4$$



Interval Notation: $(-\infty, -14] \cup [-4, \infty)$

$$11. 3|8x+5|-9 \geq 24 \quad 3|8x+5| \geq 33$$

$$|8x+5| \geq 11$$

$$8x+5 \geq 11$$

$$8x \geq 6$$

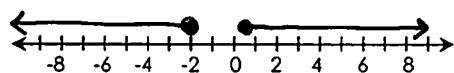
$$x \geq 0.75$$

$$8x+5 \leq -11$$

$$8x \leq -16$$

$$x \leq -2$$

$$x \leq -2 \text{ or } x \geq 0.75$$



Interval Notation: $(-\infty, -2] \cup [0.75, \infty)$

$$12. 4|9-3n|-9 < 75 \quad 4|9-3n| < 84$$

$$|9-3n| < 21$$

$$9-3n < 21$$

$$-3n < 12$$

$$n > -4$$

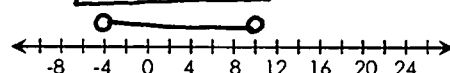
$$9-3n > -21$$

$$-3n > -30$$

$$n < 10$$

$$n > -4 \text{ and } n < 10$$

$$-4 < n < 10$$



Interval Notation: $(-4, 10)$

Name: _____

Unit 1: Equations & Inequalities

Date: _____ Per: _____

Homework 7: Absolute Value Inequalities

**** This is a 2-page document! ******Solve, graph, and write the solution to each inequality in interval notation.**

1. $|b+3| \geq 7$

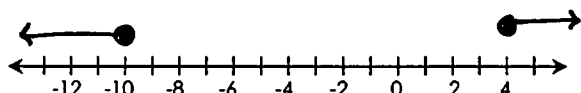
$b+3 \geq 7$

$b \geq 4$

$b+3 \leq -7$

$b \leq -10$

$b \leq -10 \text{ or } b \geq 4$

Interval Notation: $(-\infty, -10] \cup [4, \infty)$

2. $|-2v-4| < 8$

$-2v-4 < 8$

$-2v < 12$

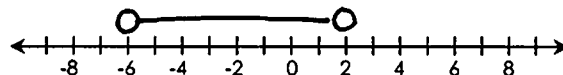
$v > -6$

$-2v-4 > -8$

$-2v > -4$

$v < 2$

$-6 < v < 2$

Interval Notation: $(-6, 2)$

3. $|5k+4| \geq 36$

$5k+4 \geq 36$

$5k \geq 32$

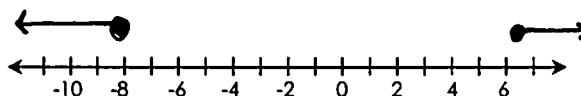
$k \geq 6.4$

$5k+4 \leq -36$

$5k \leq -40$

$k \leq -8$

$k \leq -8 \text{ or } k \geq 6.4$

Interval Notation: $(-\infty, -8] \cup [6.4, \infty)$

4. $|3-9y| \leq 33$

$3-9y \leq 33$

$-9y \leq 30$

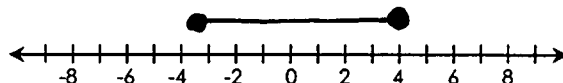
$y \geq -\frac{10}{3}$

$3-9y \geq -33$

$-9y \geq -36$

$y \leq 4$

$-\frac{10}{3} \leq y \leq 4$

Interval Notation: $[-\frac{10}{3}, 4]$

5. $|\frac{n}{7}| - 6 > -5$

$|\frac{n}{7}| > 1$

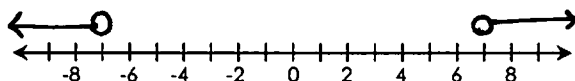
$\frac{n}{7} > 1$

$n > 7$

$\frac{n}{7} < -1$

$n < -7$

$n < -7 \text{ or } n > 7$

Interval Notation: $(-\infty, -7) \cup (7, \infty)$

6. $\frac{|5+3w|}{-2} \leq -1$

$|5+3w| \geq 2$

$5+3w \geq 2$

$3w \geq -3$

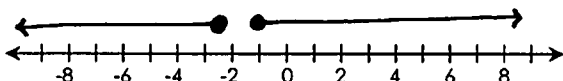
$w \geq -1$

$5+3w \leq -2$

$3w \leq -7$

$w \leq -\frac{7}{3}$

$w \leq -\frac{7}{3} \text{ or } w \geq -1$

Interval Notation: $(-\infty, -\frac{7}{3}] \cup [-1, \infty)$

7. $7|4a-6| < 42$

$$|4a-6| < 6$$

$$4a-6 < 6$$

$$4a < 12$$

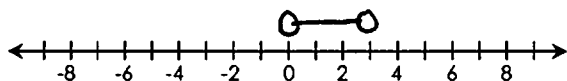
$$a < 3$$

$$4a-6 > -6$$

$$4a > 0$$

$$a > 0$$

$$0 < a < 3$$



Interval Notation: $(0, 3)$

8. $2-9|p+7| > -16$

$$-9|p+7| > -18$$

$$|p+7| < 2$$

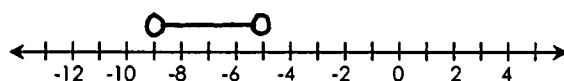
$$p+7 < 2$$

$$p < -5$$

$$p+7 > -2$$

$$p > -9$$

$$-9 < p < -5$$



Interval Notation: $(-9, -5)$

9. $3|z-4| - 13 \leq -1$

$$3|z-4| \leq 12$$

$$|z-4| \leq 4$$

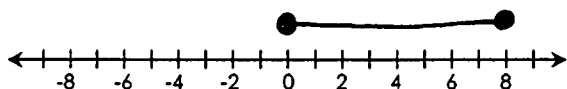
$$z-4 \leq 4$$

$$z \leq 8$$

$$z-4 \geq -4$$

$$z \geq 0$$

$$0 \leq z \leq 8$$



Interval Notation: $[0, 8]$

10. $\frac{|-2m|}{3} + 3 > 8$

$$\frac{|-2m|}{3} > 5$$

$$|-2m| > 15$$

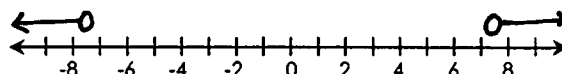
$$-2m > 15$$

$$m < -7.5$$

$$-2m < -15$$

$$m > 7.5$$

$$m < -7.5 \text{ or } m > 7.5$$



Interval Notation: $(-\infty, -7.5) \cup (7.5, \infty)$

11. $4|8y+7| - 9 < 91$

$$4|8y+7| < 100$$

$$|8y+7| < 25$$

$$8y+7 < 25$$

$$8y < 18$$

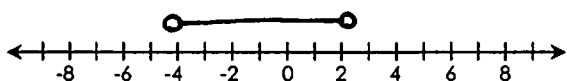
$$y < 2.25$$

$$8y+7 > -25$$

$$8y > -32$$

$$y > -4$$

$$-4 < y < 2.25$$



Interval Notation: $(-4, 2.25)$

12. $2-5|10-2n| \geq -28$

$$-5|10-2n| \geq -30$$

$$|10-2n| \leq 6$$

$$10-2n \leq 6$$

$$-2n \leq -4$$

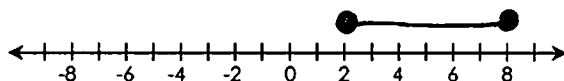
$$n \geq 2$$

$$10-2n \geq -6$$

$$-2n \geq -16$$

$$n \leq 8$$

$$2 \leq n \leq 8$$



Interval Notation: $[2, 8]$

Name: _____

Algebra 2

Date: _____ Per: _____

Unit 1: Equations & Inequalities

Quiz 1-2: Inequalities

Solve, graph, and write all final answers in interval notation.

Part I: Multi-Step Inequalities

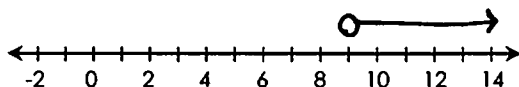
1. $5 - 4(2x - 7) < -39$

$$5 - 8x + 28 < -39$$

$$-8x + 33 < -39$$

$$-8x < -72$$

$$\boxed{x > 9}$$

Interval Notation: $(9, \infty)$

2. $11(m + 2) \leq 2m - (3m + 8)$

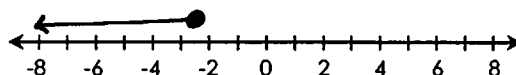
$$11m + 22 \leq 2m - 3m - 8$$

$$11m + 22 \leq -m - 8$$

$$12m + 22 \leq -8$$

$$12m \leq -30$$

$$\boxed{m \leq -2.5}$$

Interval Notation: $(-\infty, -2.5]$

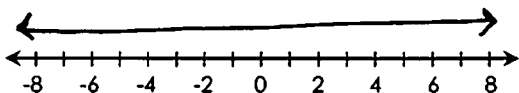
3. $2(7a - 1) - 11a \geq 3(a - 4)$

$$14a - 2 - 11a \geq 3a - 12$$

$$3a - 2 \geq 3a - 12$$

$$-2 \geq -12$$

$$\boxed{\mathbb{R}}$$

Interval Notation: $(-\infty, \infty)$

4. $\frac{5(8 - 2p) + 6}{-4} < p - 1$

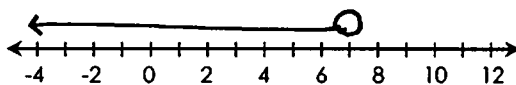
$$40 - 10p + 6 > -4p + 4$$

$$-10p + 46 > -4p + 4$$

$$-6p + 46 > 4$$

$$-6p > -42$$

$$p < 7$$

Interval Notation: $(-\infty, 7)$

Part II: Compound Inequalities

5. $\frac{3}{4}x + 7 < 1$ or $9(4 - x) \leq 36$

$$\frac{3}{4}x < -6$$

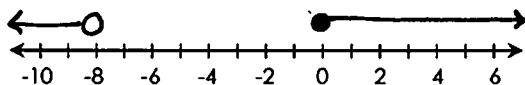
$$x < -8$$

$$36 - 9x \leq 36$$

$$-9x \leq 0$$

$$x \geq 0$$

$$\boxed{x < -8 \text{ or } x \geq 0}$$

Interval Notation: $(-\infty, -8) \cup [0, \infty)$

6. $\frac{9 - 8w}{5} > 3$ or $\frac{2}{3}(18w - 3) < -62$

$$9 - 8w > 15$$

$$-8w > 6$$

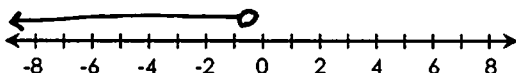
$$w < -\frac{3}{4}$$

$$12w - 2 < -62$$

$$12w < -60$$

$$w < -5$$

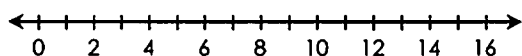
$$w < -\frac{3}{4} \text{ or } w < -5; \boxed{w < -\frac{3}{4}}$$

Interval Notation: $(-\infty, -3/4)$

7. $7 - n \geq 6$ and $2(3n - 8) \geq 20$

$$\begin{aligned} -n &\geq -1 & 6n - 16 &\geq 20 \\ n &\leq 1 & 6n &\geq 36 \\ & & n &\geq 6 \end{aligned}$$

$n \leq 1$ and $n \geq 6$; $\boxed{\emptyset}$

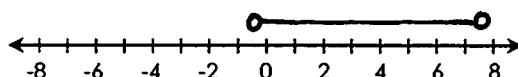


Interval Notation: $\underline{\emptyset}$

8. $-5 < 10c - 1 < 74$

$$-4 < 10c < 75$$

$$\boxed{-0.4 < c < 7.5}$$



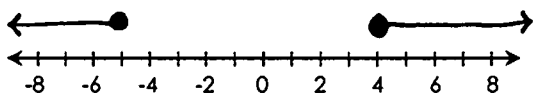
Interval Notation: $\underline{(-0.4, 7.5)}$

Part III: Absolute Value Inequalities

9. $|2k + 1| \geq 9$

$$\begin{aligned} 2k + 1 &\geq 9 & 2k + 1 &\leq -9 \\ 2k &\geq 8 & 2k &\leq -10 \\ k &\geq 4 & k &\leq -5 \end{aligned}$$

$$\boxed{k \leq -5 \text{ or } k \geq 4}$$



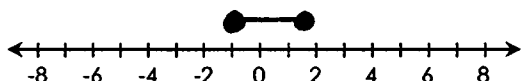
Interval Notation: $\underline{(-\infty, -5] \cup [4, \infty)}$

11. $|4m - 1| - 5 \leq 0$

$$|4m - 1| \leq 5$$

$$\begin{aligned} 4m - 1 &\leq 5 & 4m - 1 &\geq -5 \\ 4m &\leq 6 & 4m &\geq -4 \\ m &\leq 1.5 & m &\geq -1 \end{aligned}$$

$$-1 \leq m \leq 1.5$$



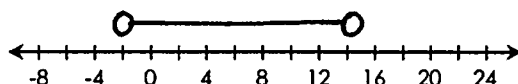
Interval Notation: $\underline{[-1, 1.5]}$

10. $-7|x - 6| > -56$

$$|x - 6| < 8$$

$$\begin{aligned} x - 6 &< 8 & x - 6 &> -8 \\ x &< 14 & x &> -2 \end{aligned}$$

$$-2 < x < 14$$



Interval Notation: $\underline{(-2, 14)}$

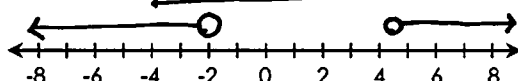
12. $10 - 4|8y - 10| < -94$

$$-4|8y - 10| < -104$$

$$|8y - 10| > 26$$

$$\begin{aligned} 8y - 10 &> 26 & 8y - 10 &< -26 \\ 8y &> 36 & 8y &< -16 \\ y &> 4.5 & y &< -2 \end{aligned}$$

$$\boxed{y < -2 \text{ or } y > 4.5}$$



Interval Notation: $\underline{(-\infty, -2) \cup (4.5, \infty)}$

Unit 1 Test Study Guide (Equations & Inequalities)

Name: _____

Date: _____ Per: _____

Topic 1: Simplifying Radicals & Classifying Numbers in the Real Number System

Simplify each radical expression.

1. $\sqrt{84}$ $= \sqrt{4 \cdot 21}$ $= \boxed{2\sqrt{21}}$	2. $\sqrt{243}$ $= \sqrt{81 \cdot 3}$ $= \boxed{9\sqrt{3}}$	3. $-\sqrt{160}$ $= -\sqrt{16 \cdot 10}$ $= \boxed{-4\sqrt{10}}$	4. $\sqrt{450}$ $= \sqrt{225 \cdot 2}$ $= \boxed{15\sqrt{2}}$
---	---	--	---

List ALL sets to which each value belongs. (Use R, I, Q, Z, W, N)

5. $\frac{\sqrt{3}}{\sqrt{3}} = 1$ R, Q, Z, W, N	6. $5.08\bar{3}$ R, Q	7. $\frac{-\sqrt{256}}{2} = \frac{-16}{2} = -8$ R, Q, Z	8. 2π R, I
---	--------------------------	--	-------------------

Topic 2: Operations & Expressions

Simplify each expression.

9. $[18 - (-1-7)^2] + 16 \div 2^4$ $= [18 - (-8)^2] + 16 \div 16$ $= [18 - 64] + 1$ $= -46 + 1$ $= \boxed{-45}$	10. $\frac{4 + \sqrt{121} - 2 \cdot 3^3}{ -19 - 2(-8) } = \frac{4 + 11 - 2 \cdot 27}{ -19 + 16 }$ $= \frac{15 - 54}{ -3 }$ $= \frac{-39}{3} = \boxed{-13}$
11. $10a - 5ab + 4b$ (if $a = \frac{2}{5}$ and $b = -\frac{1}{6}$) $= 10(\frac{2}{5}) - 5(\frac{2}{5})(-\frac{1}{6}) + 4(-\frac{1}{6})$ $= 4 + \frac{1}{3} - \frac{2}{3}$ $= \frac{12}{3} + \frac{1}{3} - \frac{2}{3} = \boxed{\frac{11}{3}}$	12. $\sqrt{-x^2 - 4y^2}$ (if $x = 3$ and $y = -2$) $= \sqrt{-(3)^2 - 4(-2)^2}$ $= \sqrt{-9 - 4(4)}$ $= \sqrt{-9 - 16}$ $= \sqrt{-25} = \sqrt{25} = \boxed{5}$

Topic 3: Multi-Step Equations & Word Problems

Solve each equation.

13. $3(7 - 9k) + 23k = 4k - (24 - k)$ $21 - 27k + 23k = 4k - 24 + k$ $21 - 4k = 5k - 24$ $21 = 9k - 24$ $45 = 9k$ $\boxed{5 = k}$	14. $7 - \frac{5}{2}(8n - 18) = 14 - 10(2n - 3)$ $7 - 20n + 45 = 14 - 20n + 30$ $-20n + 52 = -20n + 44$ $52 \neq 44$ $\boxed{\emptyset}$
--	--

15. $\frac{6x-11}{7} = \frac{2x-9}{3}$ $3(6x-11) = 7(2x-9)$ $18x - 33 = 14x - 63$ $4x - 33 = -63$ $4x = -30$ $x = -7.5$	16. If $SA = \frac{1}{2}lp + B$, find p $SA - B = \frac{1}{2}lp$ $2(SA - B) = lp$ $\frac{2(SA - B)}{l} = p$
17. The width of a rectangle is four less than one half its length. If the perimeter of the rectangle is 94 meters, find the area of the rectangle. let $x = \text{length}$ let $\frac{1}{2}x - 4 = \text{width}$ $2(x) + 2(\frac{1}{2}x - 4) = 94$ $2x + x - 8 = 94$ $3x = 102$ $x = 34 \rightarrow \text{length} = 34$ $\text{width} = 13$ $\text{Area} = 34(13) = 442 \text{ m}^2$	18. Find three consecutive odd numbers such that the sum of five times the smaller number and twice the larger number is 33 more than six times the median number. let $x = 1^{\text{st}} \text{ odd int.}$ let $x+2 = 2^{\text{nd}} \text{ odd int.}$ let $x+4 = 3^{\text{rd}} \text{ odd int.}$ $5(x) + 2(x+4) = 6(x+2) + 33$ $5x + 2x + 8 = 6x + 12 + 33$ $7x + 8 = 6x + 45$ $x + 8 = 45$ $x = 37$ $37, 39, 41$

Topic 4: Absolute Value Equations

Solve each equation. Be sure to check for extraneous solutions.

19. $-7-4x =17$ $-7-4x=17 \quad -7-4x=-17$ $-4x=24 \quad -4x=-10$ $x=-6 \checkmark \quad x=2.5 \checkmark$ $x = \{-6, 2.5\}$	20. $\frac{ 5n-10 }{-2} = -15$ $5n-10 = 30$ $5n-10=30 \quad 5n-10=-30$ $5n=40 \quad 5n=-20$ $n=8 \checkmark \quad n=-4 \checkmark$ $n = \{-4, 8\}$
21. $2-10 k+1 =-78$ $-10 k+1 =-80$ $ k+1 =8$ $k+1=8 \quad k+1=-8$ $k=7 \checkmark \quad k=-9 \checkmark$ $k = \{-9, 7\}$	22. $2m+7 =6m+13$ $2m+7=6m+13 \quad 2m+7=-6m-13$ $-4m+7=13 \quad 8m+7=-13$ $-4m=6 \quad 8m=-20$ $m=-1.5 \checkmark \quad m=-2.5$ $m = \{-1.5\}$

Topic 5: Multi-Step Inequalities

Solve and graph each inequality. Write your answer in interval notation.

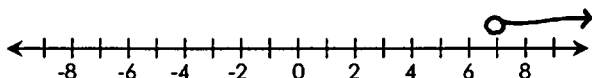
23. $33x - 8(3x + 9) > -9$

$$33x - 24x - 72 > -9$$

$$9x - 72 > -9$$

$$9x > 63$$

$$\boxed{x > 7}$$



Interval Notation: $(7, \infty)$

24. $-92 - 8k \geq 23 - (7 - 4k)$

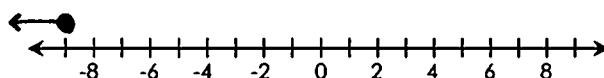
$$-92 - 8k \geq 23 - 7 + 4k$$

$$-92 - 8k \geq 16 + 4k$$

$$-92 - 12k \geq 16$$

$$-12k \geq 108$$

$$\boxed{k \leq -9}$$



Interval Notation: $(-\infty, -9]$

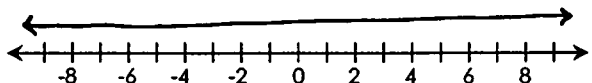
25. $11(5u - 4) - 7u \geq 8(6u - 7)$

$$55u - 44 - 7u \geq 48u - 56$$

$$48u - 44 \geq 48u - 56$$

$$-44 \geq -56$$

$$\boxed{\mathbb{R}}$$



Interval Notation: $(-\infty, \infty)$

26. $-\frac{5}{3}\left(\frac{9}{10}x + 15\right) < 7 - \left(8 - \frac{9}{2}x\right)$

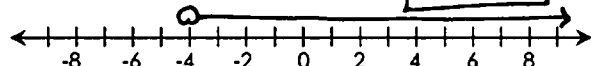
$$-\frac{3}{2}x - 25 < 7 - 8 + \frac{9}{2}x$$

$$-\frac{3}{2}x - 25 < -1 + \frac{9}{2}x$$

$$-25 < -1 + 6x$$

$$-24 < 6x$$

$$\boxed{x > -4}$$



Interval Notation: $(-4, \infty)$

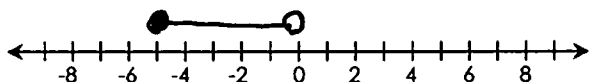
Topic 6: Compound Inequalities

Solve and graph each compound inequality. Write your answer in interval notation.

27. $-18 \leq 2b - 8 < -8$

$$-10 \leq 2b < 0$$

$$-5 \leq b < 0$$



Interval Notation: $[-5, 0)$

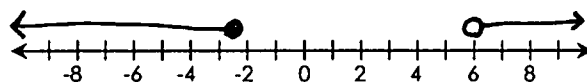
28. $m + 5 > 11$ or $8 - 10m \geq 33$

$$m > 6$$

$$-10m \geq 25$$

$$m \leq -2.5$$

$$\boxed{m \leq -2.5 \text{ or } m > 6}$$



Interval Notation: $(-\infty, -2.5] \cup (6, \infty)$

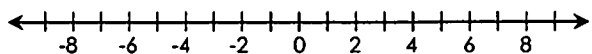
29. $7p + 5 \leq -37$ and $-10p < 10$

$$7p \leq -42$$

$$p > -1$$

$$p \leq -6$$

$$p < -6 \text{ and } p > -1 ; \boxed{\emptyset}$$



Interval Notation: $\emptyset$

30. $7 - 3x \leq -20$ or $5x - 6 \leq 9$

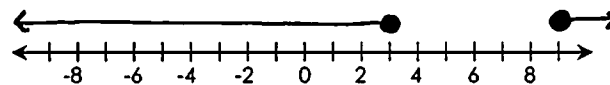
$$-3x \leq -27$$

$$5x \leq 15$$

$$x \geq 9$$

$$x \leq 3$$

$$\boxed{x \leq 3 \text{ or } x \geq 9}$$



Interval Notation: $(-\infty, 3] \cup [9, \infty)$

31. $9y - 2 < 34$ and $3y - 2 > -29$

$$9y < 36$$

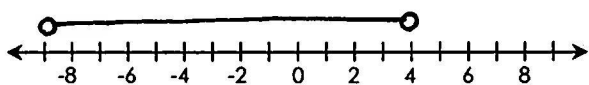
$$y < 4$$

$$3y > -27$$

$$y > -9$$

$$y > -9 \text{ and } y < 4$$

$$\boxed{-9 < y < 4}$$



Interval Notation: $(-9, 4)$

32. $10 + 2w \geq 30$ or $5w - 8 > -13$

$$2w \geq 20$$

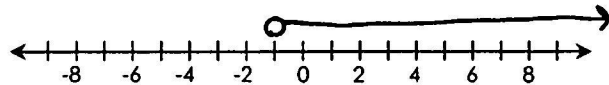
$$5w > -5$$

$$w \geq 10$$

$$w > -1$$

$$w > -1 \text{ or } w \geq 10;$$

$$\boxed{w > -1}$$



Interval Notation: $(-1, \infty)$

Topic 7: Absolute Value Inequalities

Solve and graph each absolute value inequality. Write your answer in interval notation.

33. $|7 - a| \geq 2$

$$7 - a \geq 2$$

$$7 - a \leq -2$$

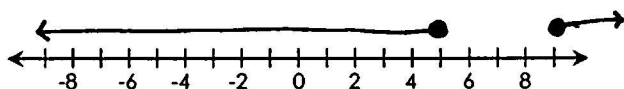
$$-a \geq -5$$

$$-a \leq -9$$

$$a \leq 5$$

$$a \geq 9$$

$$\boxed{a \leq 5 \text{ or } a \geq 9}$$



Interval Notation: $(-\infty, 5] \cup [9, \infty)$

34. $\frac{|v-3|}{-5} > -1$

$$|v-3| < 5$$

$$v-3 < 5$$

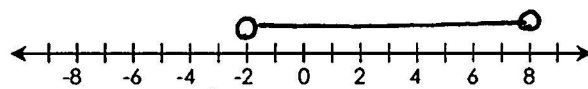
$$v-3 > -5$$

$$v < 8$$

$$v > -2$$

$$v > -2 \text{ and } v < 8$$

$$\boxed{-2 < v < 8}$$



Interval Notation: $(-2, 8)$

35. $|3n+8|+1 > 5$

$$|3n+8| > 4$$

$$3n+8 > 4$$

$$3n+8 < -4$$

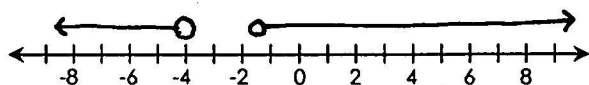
$$3n > -4$$

$$3n < -12$$

$$n > -\frac{4}{3}$$

$$n < -4$$

$$\boxed{n < -4 \text{ or } n > -\frac{4}{3}}$$



Interval Notation: $(-\infty, -4) \cup (-\frac{4}{3}, \infty)$

36. $-3 - 6|4x-10| < -87$

$$-6|4x-10| < -84$$

$$|4x-10| > 14$$

$$4x-10 > 14$$

$$4x-10 < -14$$

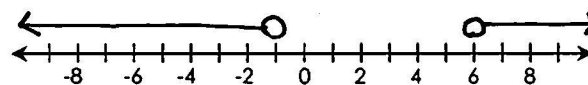
$$4x > 24$$

$$4x < -4$$

$$x > 6$$

$$x < -1$$

$$\boxed{x < -1 \text{ or } x > 6}$$



Interval Notation: $(-\infty, -1) \cup (6, \infty)$

Name: _____

Date: _____ Per: _____

Unit 1 Test

Equations & Inequalities

1. Which value is a perfect square?

A. 198

B. 249

C. 361 — 19 · 19

D. 500

C

2. If the expression below is written in simplest radical form, which must be true?

$$\sqrt{208} = \boxed{a} \sqrt{\boxed{b}}$$

$$\sqrt{208} = \sqrt{16 \cdot 13} \\ = 4\sqrt{13}$$

A. $a + b = 17$ B. $a + b = 23$ C. $a + b = 29$ D. $a + b = 31$

A

3. Which expression is correctly written in simplest form?

A. $\sqrt{128} = 4\sqrt{8}$

B. $\sqrt{180} = 6\sqrt{5}$

C. $\sqrt{72} = 12\sqrt{3}$

D. $\sqrt{28} = 4\sqrt{7}$

B

4. Which set of numbers does **not** contain the value below?

$$-\frac{\sqrt{64}}{2} = -\frac{8}{2} = -4$$

A. R B. N C. Z D. Q

B

5. Which of the following lists of values contain rational numbers only?

A. 90, -1.375, π

C. $\frac{63}{9}$, $\sqrt{\frac{1}{16}}$, $\sqrt{50}$

B. $\sqrt{28}$, 0, $3\frac{1}{5}$

D. $-\sqrt{81}$, $\frac{\sqrt{7}}{\sqrt{7}}$, $4.\overline{16}$

D

6. Which is the smallest subset of the real numbers that contains the value below?

$$\sqrt{3} - \sqrt{3} = 0$$

A. R B. N C. W D. Z

C

7. Simplify the expression below.

$$\begin{aligned} &\sqrt{324} - (2 - 9)^2 + 162 \div 3^3 \\ &18 - (-7)^2 + 162 \div 27 \\ &18 - 49 + 6 \\ &-31 + 6 \end{aligned}$$

-25

8. Simplify the expression below.

$$\begin{aligned} &\frac{\sqrt{2^6} + (11 - 5^2)}{|-10 - 2(3)|} \\ &\frac{\sqrt{64} + (11 - 25)}{|-10 - 6|} = \frac{8 + (-14)}{|-16|} \\ &= \frac{-6}{16} \end{aligned}$$

-3/8

9. Evaluate the expression below if $m = -4$, $n = 3$, and $p = 12$.

$$\frac{m^2n + 9p}{\sqrt{np}}$$

$$= \frac{(-4)^2(3) + 9(12)}{\sqrt{3(12)}}$$

$$= \frac{16(3) + 108}{\sqrt{36}}$$

$$= \frac{156}{6} = 26$$

26

10. Evaluate the expression below if $x = \frac{4}{9}$, $y = \frac{1}{2}$, and $z = \frac{3}{5}$.

$$6xy - 3z$$

$$= 6\left(\frac{4}{9}\right)\left(\frac{1}{2}\right) - 3\left(\frac{3}{5}\right)$$

$$= \frac{24}{18} - \frac{9}{5}$$

$$= -\frac{7}{15}$$

$-\frac{7}{15}$

11. Solve the equation below.

$$8a - (1 + 5a) = 4(3a - 11) - 20$$

$$8a - 1 - 5a = 12a - 44 - 20$$

$$3a - 1 = 12a - 64$$

$$63 = 9a$$

$$a = 7$$

$a = 7$

12. Solve the equation below.

$$8 - \frac{4}{3}\left(6 - \frac{5}{8}x\right) = \frac{1}{2}\left(\frac{7}{3}x + 18\right)$$

$$8 - 8 + \frac{5}{6}x = \frac{7}{6}x + 9$$

$$\frac{5}{6}x = \frac{7}{6}x + 9$$

$$-\frac{1}{3}x = 9$$

$$x = -27$$

$x = -27$

13. Which equation has a solution of all real numbers?

- A. $5k - (6k - 18) = 9 - k + 17$
 B. $14 - 5(y + 10) = 4 - 8(y + 5)$
 C. $6(-2w - 3) = -4(3w + 7) + 10$
 D. $7m - 2(2m + 3) = 8 - (3m - 10)$

✓ $-12w - 18 = -12w - 28 + 10$
 $-12w - 18 = -12w - 18$

C

14. Solve the equation below.

$$\frac{5p - 8}{2} = \frac{7p + 4}{2}$$

$$10p - 16 = 14p + 8$$

$$-16 = 4p + 8$$

$$-24 = 4p$$

$$-6 = p$$

$p = -6$

15. Solve the equation below for m .

$$\frac{2m-n}{5} = 7n$$

$$2m-n = 35n$$

$$2m = 36n$$

$$m = 18n$$

A. $m = 40n$

B. $m = 18n$

C. $m = 8n$

D. $m = 20n$

B

16. If $2x - 1 = 13$, evaluate the expression below.

$$\begin{aligned} &\downarrow \\ 2x &= 14 \\ x &= 7 \end{aligned}$$

$$\begin{aligned} &-x^2 - 4x \\ &-(7)^2 - 4(7) \\ &-49 - 28 \\ &= -77 \end{aligned}$$

A. 12

B. 21

C. -60

D. -77

D

17. This past Sunday, the Giants scored 9 points less than twice the Cowboys. The Packers scored 14 more points than the Giants. If the teams scored 81 total points, find the number of points scored by the Packers.

let x = Cowboys
let $2x-9$ = Giants
let $2x+5$ = Packers

$$\begin{aligned} x + 2x - 9 + 2x + 5 &= 81 \\ 5x - 4 &= 81 \\ 5x &= 85 \\ x &= 17 \end{aligned}$$

Packers: $2(17) + 5 = 39$

39

18. The sum of three consecutive numbers is 41 less than five times the largest number. Find the largest number.

let x = 1st consec. #
 $x+1$ = 2nd consec. #
 $x+2$ = 3rd consec. #

$$\begin{aligned} x + x + 1 + x + 2 &= 5(x+2) - 41 \\ 3x + 3 &= 5x + 10 - 41 \\ 3x + 3 &= 5x - 31 \\ 34 &= 2x \\ x &= 17 \end{aligned}$$

largest = $17 + 2 = 19$

19

19. Solve the equation below.

$$|3x - 9| = 24$$

$$\begin{aligned} 3x - 9 &= 24 & 3x - 9 &= -24 \\ 3x &= 33 & 3x &= -15 \\ x &= 11 & x &= -5 \end{aligned}$$

$x = \{-5, 11\}$

20. Solve the equation below.

$$\begin{aligned} -2|7 - 9c| + 7 &= -15 \\ -2|7 - 9c| &= -22 \\ |7 - 9c| &= 11 \end{aligned}$$

$$\begin{aligned} 7 - 9c &= 11 & 7 - 9c &= -11 \\ -9c &= 4 & -9c &= -18 \\ c &= -\frac{4}{9} & c &= 2 \end{aligned}$$

$c = \{-\frac{4}{9}, 2\}$

21. Solve the equation below.

$$|6w + 4| = 2w - 10$$

$$6w + 4 = 2w - 10$$

$$4w + 4 = -10$$

$$4w = -14$$

$$w = -\frac{7}{2}$$

$$6w + 4 = -2w + 10$$

$$8w + 4 = 10$$

$$8w = 6$$

$$w = \frac{3}{4}$$

$$w = \emptyset$$

22. Which graph represents the solution to the inequality below?

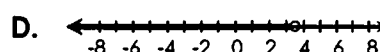
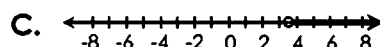
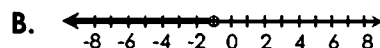
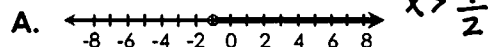
$$\frac{5 - (9 - 4x)}{-2} < -5$$

$$5 - 9 + 4x > 10$$

$$-4 + 4x > 10$$

$$4x > 14$$

$$x > \frac{7}{2}$$



C

23. Which represents the solution to the inequality below in interval notation?

$$-\frac{1}{3}(21 - 9n) \geq 4(2n - 7) - 9$$

$$-7 + 3n \geq 8n - 28 - 9$$

$$-7 + 3n \geq 8n - 37$$

$$-7 \geq 5n - 37$$

$$30 \geq 5n$$

A. $(-\infty, 6]$

$$6 \geq n$$

B. $(-\infty, -6]$

$$n \leq 6$$

C. $[6, \infty)$

D. $[-6, \infty)$

A

24. Which graph represents the solution to the compound inequality below?

$$-4k - 3 < -7 \text{ or } 1 - k \geq 6$$

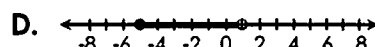
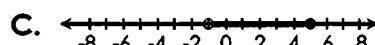
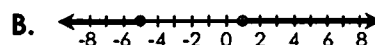
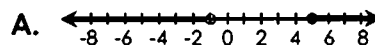
$$-4k < -4$$

$$-k \geq 5$$

$$k > 1$$

$$k \leq -5$$

$$k \leq -5 \text{ or } k > 1$$



B

25. Which graph represents the solution to the compound inequality below?

$$-1 + 9y > -73 \text{ and } \frac{y - 4}{-3} > 2$$

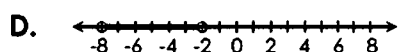
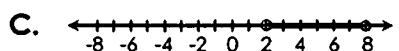
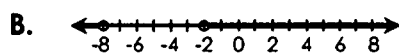
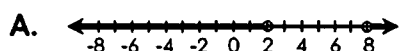
$$9y > -72$$

$$y - 4 < -6$$

$$y > -8$$

$$y < -2$$

$$-8 < y < -2$$



D

26. Which graph represents the solution to the compound inequality below?

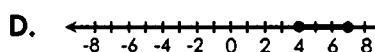
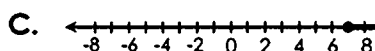
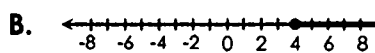
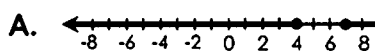
$$\frac{1}{2}a - 5 \geq -3 \text{ or } -6a - 14 \leq -56$$

$$\frac{1}{2}a \geq 2$$

$$-6a \leq -42$$

$$a \geq 4$$

$$a \geq 7$$



B

27. Which represents the solution to the compound inequality below in interval notation?

$$5 - 2x > 31 \text{ and } 7x + 8 \geq -55$$

$$-2x > 26 \qquad 7x \geq -63$$

$$x < -13 \qquad x \geq -9$$

- A. $(-13, -9]$
- B. $(-\infty, -13) \cup [-9, \infty)$
- C. $[-9, \infty)$
- D. $\emptyset$

D

28. Which value of x is **not** in the solution to the compound inequality below?

$$-15 \leq 2x - 9 < 5$$

$$-6 \leq 2x < 14$$

$$-3 \leq x < 7$$

- A. -3
- B. 0
- C. 4
- D. 7

D

29. Which graph represents the solution to the inequality below?

$$|9x + 3| \leq 21$$

$$9x + 3 \leq 21$$

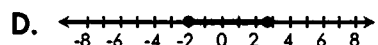
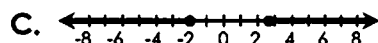
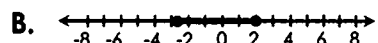
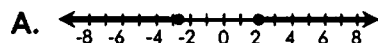
$$9x \leq 18$$

$$x \leq 2$$

$$9x + 3 \geq -21$$

$$9x \geq -24$$

$$x \geq -\frac{8}{3}$$



B

30. Which represents the solution to the inequality below in interval notation?

$$-3|n - 5| - 11 < -38$$

$$-3|n - 5| < -27$$

$$|n - 5| > 9$$

$$n - 5 > 9$$

$$n > 14$$

$$n - 5 < -9$$

$$n < -4$$

- A. $(-4, 14)$
- B. $(-14, 4)$
- C. $(-\infty, -4) \cup (14, \infty)$
- D. $(-\infty, -14) \cup (4, \infty)$

C

BONUS: A change jar contains nickels, dimes, and quarters, totaling an amount of \$1.90. The number of nickels is one more than twice the number of dimes. The number of quarters is one half the total number of nickels and dimes. Find the number of each coin in the change jar.

$$\text{let } x = \text{dimes}$$

$$\text{let } 2x + 1 = \text{nickels}$$

$$\text{let } \frac{1}{2}(3x + 1) = \text{quarters}$$

$$.10x + .05(2x + 1) + .25(\frac{1}{2}(3x + 1)) = 1.90$$

$$.10x + .10x + .05 + .375x + .125 = 1.90$$

$$.575x + .175 = 1.90$$

$$.575x = 1.725$$

$$x = 3$$

$$\text{dimes} = 3$$

$$\text{nickels} = 2(3) + 1 = 7$$

$$\text{quarters} = \frac{1}{2}(3(3) + 1) = 5$$

3 dimes, 7 nickels, 5 quarters

CREDITS

I use clipart and
fonts in my products by:



Art with Jenny K



Many thanks to these
talented artists!